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FACULTY OF SOCIAL SCIENCES

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**Forecasting Tourism Demand Using
Artificial Neural Networks**

Master's thesis

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Declaration of Authorship

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Prague, July 28, 2025

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Abstract

Tourism plays a vital role in Croatia's economy, yet accurately forecasting demand remains a complex topic, especially during periods of economic volatility. This thesis explores the use of artificial neural networks (ANN) to forecast monthly tourist arrivals and overnights from 23 origin countries, using data spanning from 2000 to 2024. The modeling framework incorporates macroeconomic indicators, policy-related variables, such as EU, Schengen and euro adoption, Google Trends data, and autoregressive lags, with country-level embeddings in order to account for heterogeneity across different markets. To assess how models respond to shocks, forecasts were estimated both with and without data from the COVID-19 period. The results show that in stable periods, demand is mainly driven by seasonality and recent trends, which explains the solid performance of traditional autoregressive models. However, when the data includes periods of disruption, neural networks capture broader patterns and outperform the benchmark. In such instances, macroeconomic and behavioral signals also become slightly more relevant in forecasting tourism demand. While overall performance is strong, the ANN tends to underpredict high-volume markets during recovery years. A targeted retraining approach leads to moderate improvements in these cases. The results also show that non-Schengen countries are more sensitive to short-term shifts, likely due to visa regimes and planning behavior. Overall, the findings demonstrate that neural networks offer a flexible and robust forecasting framework, capable of capturing both stable and volatile demand patterns, supporting more responsive tourism planning both during times of crisis and surrounded by pressures of overtourism. While certain limitations remain, the findings highlight the advantages of flexible models like neural networks in capturing both stable patterns and sudden shifts driven by external disruptions in tourism demand.

JEL Classification Z32, Z38

Keywords Tourism demand, Artificial Neural Networks, Forecasting, Croatia

Title Forecasting Tourism Demand Using Artificial Neural Networks

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Acronyms

ADF	Augmented Dickey-Fuller test
ACF	Autocorrelation Function
ANN	Artificial Neural Network
AR	Autoregressive
ARIMA	Autoregressive Integrated Moving Average
AI	Artificial Intelligence
ML	Machine Learning
DL	Deep Learning
FFNN	Feedforward Neural Network
ReLU	Rectified Linear Unit
MSE	Mean Squared Error
RMSE	Root Mean Squared Error
MAE	Mean Absolute Error
MAPE	Mean Absolute Percentage Error
SGD	Stochastic Gradient Descent
CPI	Consumer Price Index
OECD	Organisation for Economic Co-operation and Development
UNWTO	United Nations World Tourism Organization
EU	European Union
STL	Seasonal-Trend Decomposition using LOESS
EC	European Commission
ECB	European Central Bank

Master's Thesis Proposal

Author	Bc. Aida Hodzic
Supervisor	prof. PhDr. Ladislav Kristoufek, Ph.D.
Proposed topic	Forecasting Tourism Demand Using Artificial Neural Networks

Motivation Tourism sector is one of the fastest-growing industries that powers the global economic growth, and a major contributor to globalization and the increasing connectedness of the world. (World Economic Forum, 2024)

The importance of understanding tourism determinants and being able to deliver accurate takeaways about its demand lies in the fact that the tourism sector is a key driver of economic prosperity for many countries through its impact on job creation, improvement of balance of payments, infrastructure developments, international relations and worldwide business opportunities. (OECD, 2024) UNWTO defines it as the “social, cultural and economic phenomenon which entails the movement of people to countries or places outside their usual environment for personal or business/professional purposes.” (UNWTO, 2022).

The main motivation behind this work is to develop a forecasting model that can support better decision-making in the tourism sector, while acknowledging the challenges and limitations involved. The key contribution of the thesis lies in exploring a broad set of factors that can influence tourism demand, such as currency regime, spoken language, geographic location, price levels, transport accessibility, and institutional ties like European Union membership, entry to the Schengen Area and Eurozone. The model is applied to Croatia, a country that recently adopted the euro and joined the Schengen Area, allowing for a comparison of demand patterns before and after these policy shifts. While the analysis is focused on Croatia, the findings are intended to offer insights relevant to other destinations as well, particularly in contexts where tourism policy must balance the goals of growth, seasonality management, and concerns around overtourism.

Hypotheses

Hypothesis #1: Tourism demand in Croatia is shaped by a combination of seasonality, prior travel patterns, macroeconomic conditions, and broader structural factors.

Hypothesis #2: Integration into the Schengen Area may act as a relevant driver of tourism demand, particularly during periods of economic or geopolitical uncertainty, or for travelers from countries where EU membership enhances perceptions of accessibility, safety, or administrative simplicity.

Hypothesis #3: Using the euro could matter for tourism demand, particularly in situations where stable prices or easier transactions become more important — for example, during or after a crisis.

Methodology In order to make a successful analysis, the thesis employs a dataset consisting of monthly data spanning from 2001 to 2024. Tourism demand is measured by the number of registered overnight stays in Croatia, across a sample of around 20 different origin countries. In the employed Neural Network analysis, this dependent variable represents the output layer. The data is sourced from stays registered in Croatia's official administrative system, eVisitor. Independent variables were collected from multiple sources, including the Croatian Bureau of Statistics, Eurostat, the National Bank of Croatia, and the World Bank Development Indicators. These make up the input layer of the ANN model and include log-transformed macroeconomic indicators such as exchange rates, consumer price index (CPI), and unemployment rates, along with country-level policy indicators like Schengen Area membership and euro adoption.

Neural Networks typically include hidden layers, which vary depending on the complexity of the problem. Simply explained, they are made up of small units called neurons that process input data passed from the input layer (M. Uzair & N. Jamil, 2020). Choosing the right number of hidden layers requires experimentation to avoid overfitting the model (T. Gupta & K. Raza, 2018). The model selection process relies on mean squared error (MSE) as the main performance metric.

In the hidden layers of Artificial Neural Networks, activation functions help capture non-linear relationships between the input and output layers (S. Sharma & S. Sharma, 2020). This analysis uses tanh activation functions for their balanced performance in practice. In addition, the models are trained using standard deep learning hyperparameters such as learning rate, batch size, number of epochs, and optimizer settings. The final model configuration was selected based on validation set performance.

The data is split into training, validation, and test sets based on a time-based cut-off, and the final model's accuracy is assessed using MSE, RMSE, MAE, MAPE, and

R^2 . The analysis also applies permutation feature importance to better understand which variables contribute most to model performance. After identifying the key features, the thesis explores the relationships driving tourism demand and whether EU integration and the adoption of the euro have had a noticeable effect. Robustness checks include using alternative specifications and outcome variables such as tourist arrivals. The aim is to assess whether Neural Networks offer improved predictive performance compared to traditional forecasting tools — particularly in periods of policy change or tourism uncertainty, such as the COVID-19 pandemic or growing concerns around overtourism.

Expected Contribution By working on this topic, I plan to contribute to the current literature by finding the most significant factors that influence tourism demand in Croatia and hopefully deliver takeaways that can be useful for optimal decision-making and passed onto tourism demand forecasting in other countries as well. By relying on the Neural Network model as means of obtaining the results, I aim to deliver accurate results that can be useful to a wide range of businesses, governments and relevant authorities. Not only can it be beneficial for businesses directly involved in the tourism industry, such as tourist agencies, but also the ones in the field of transportation, hospitality, real estate, event planning and retail. (European Travel Commission, 2021) Having a clear understanding of the tourism demand movements can assist these stakeholders in maximizing their operational efficiency by being able to plan accordingly their strategies on staffing, inventory and marketing activities. Governments and relevant authorities can benefit from the results by including the tourism demand forecasts in their future budget allocations, investment plans, as well as decisions on country's fiscal policies, trade agreements and visa regulations that can be adjusted towards the desired level of tourism demand in the country. (Menchenko A., Javed M., 2023)

Outline

1. Motivation: This thesis aim to address a gap in the literature by applying neural network-based forecasting to the case of Croatia, a country heavily reliant on tourism and recently integrated into the Schengen Area and Eurozone. The gap is further tackled by conducting a per-country analysis to capture differences across origin markets.
2. Data: I will explain the detailed process of data collection from the listed sources and the collected list of variables.
3. Methods: I will explain the employed artificial neural network and its layers in detail, along with the measures of models' performance and accuracy.

4. Results: I will present the results on the behavior of variables concluded from the analysis and discuss if the results align with the theoretical expectations and other existing academic literature.
5. Concluding remarks: I will summarize my findings, along with the future lines of research.

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Chapter 1

Introduction

Tourism is defined as an industry that cultivates economic development and contributes significantly to job creation, international cooperation and infrastructure investments across countries. According to the Organization for Economic Cooperation and Development OECD (2022), tourism sector is the key driver of economic prosperity for many countries through its impact on job creation, improvement of balance of payments and worldwide business opportunities. Understanding tourism demand and its determinants is essential for making informed decisions that optimize the sector's growth. This is particularly relevant for Croatia, a country where tourism currently accounts for nearly 20% of the GDP, as mentioned by Orsini & Ostojic (2018). Tourist industry in Croatia is growing rapidly, with 20.93 million registered tourists arrivals and 107.3 million of tourist overnight stays in 2024, according to Croatian National Tourist Board (2024).

Despite its economic benefits, the magnitude at which tourism is growing in the region raises important questions about long-term sustainability. According to Mesas (2025), popular destinations such as Dubrovnik have reported more than 10,000 daily visitors, which is straining Croatia's infrastructure and affecting local communities. The Adriatic Sea is also faced with growing threats of plastic pollution and overfishing, while national parks are exposed to risks such as illegal construction. This further emphasizes the need for accurate forecasting of tourism demand in order to help balance the economic development with the protection of natural and cultural resources through informed decision making and long-term policy planning.

Uncovering the complex relationships between key determinants of tourism demand is just as important as the ability to forecast demand itself, as they

both contribute to understanding the different mechanisms behind tourism behavior. Therefore this thesis aims to uncover and understand the main drivers of tourism demand in Croatia, with a particular focus on how their relative importance evolves across different economic conditions and countries of origin.

In this thesis, Artificial Neural Networks (ANNs) are employed as the primary method for forecasting tourism demand in Croatia, as they are particularly useful in capturing the nonlinear relationships between economic variables and tourism flow, which traditional linear models might miss. ANNs offer a more accurate prediction of tourism trends, due to their ability to learn from high-volume, historical data and identify complex patterns. These findings can assist policymakers, businesses and stakeholders to gain a better understanding of the factors that drive tourism flows and help in optimizing resource allocation and strategies to stimulate economic growth in ways that are aligned with environmental preservation and protection of local communities.

The contribution of this thesis is in its ability to provide both theoretical and practical insights into tourism demand. While numerous literature exists on forecasting tourism demand using artificial neural networks, this thesis builds on those studies as foundation to validate the chosen methodology and apply a comprehensive modeling approach that integrates a wide range of inputs, including macroeconomic indicators, policy changes, seasonal patterns and recent trends across a 25-year period marked by both stability and disruption caused by the COVID-19 pandemic. This thesis aims to demonstrate how forecasting models can adapt to evolving conditions and offer valuable insights for countries facing similar transitions and rising tourism pressures.

Following this chapter, a review of literature focused on tourism demand forecasting and relevant approaches is presented in Chapter 2, with a particular focus on artificial neural networks and the justification for the selected approach and included variables. Chapter 3 outlines the methodological framework, including the model structures, input features, hyperparameter tuning and evaluating the performance metrics compared to a benchmark model. Chapter 4 describes the dataset, sources, and steps taken to prepare the data for modeling. Chapter 5 presents an analysis of forecasting performances of different models and countries of origin and presents the key determinants of tourism demand. The summary of the derived findings, along with a discussion on their implications, contributions and future recommendations is presented in the final Chapter 6.

Chapter 2

Literature Review

This chapter will offer a discussion of relevant studies that the thesis relies on as supporting evidence for the choice of included variables, as well as the justification for the chosen approaches in the scope of tourism forecasting and examination of underlying relationships between tourism demand and its determinants. There have been multiple studies that succeeded in its efforts to model and forecast tourism demand, but the highest accuracy seems to be achieved when deep learning methods are employed.

2.1 Artificial Neural Networks in Tourism Data

There has been a large number of published literature in the recent years that relies on artificial neural networks to successfully forecast tourism demand with high accuracy. Charef & Ayachi (2016) in their research claim that forecasting time series is undoubtedly better achieved by the artificial neural network approach, as its robustness and efficiency make it a superior choice compared to traditional forecasting models based on conditional volatility. Research by Mitrea *et al.* (2009); Dhamija & Bhalla (2010); Pao (2008) presents results that further support neural networks performing better in modeling non-linearity and complex relationships than traditional forecasting methods, primarily due to its ability to bypass the need for specifying functional forms in order to estimate parameters. Utilizing neural networks in the specific case of forecasting tourism demand is supported by numerous studies as well, including Claveria *et al.* (2015), that argues the flexible structure of neural networks and their independence of any assumptions makes them a particularly useful tool in modeling tourism data. This is explained by a typical strong presence of seasonality

and volatility within tourism data and the capability of neural networks to perform well in such situations. Similar findings have been presented in numerous research papers and articles that further provide evidence that supports using neural networks as a methodological approach in this thesis, including Constantino *et al.* (2016), Tuncsiper (2023), Teixeira & Fernandes (2014) that base their analysis on Mozambique, Türkiye and Portugal respectively. Chang-Jui *et al.* (2011) in their study conclude that the artificial neural network model generates the best results in predicting the number of tourist arrivals in Taiwan, when compared to models employing multi-variable regression.

Due to the numerous existing literature that supports utilizing ANN models in such scenarios, the thesis will use the same approach to generate results and attempt to answering the research questions with clarity and accuracy, while exploring the relationships between the tourism demand determinants in Croatia.

2.2 Tourism Demand Determinants

Understanding and predicting tourist behavior can only be done by examining tourism demand, which Frechtling (2001) describes as the extent to which tourists use goods and services. According to Morley (1992) and Schulmeister (1982), although derived from the classical economic understanding of demand, the tourism demand consists of a combination of complementary goods and services, where the subject of trade are not goods and services, but rather the consumers themselves.

Majority of the published literature uses the number of tourist arrivals to the observed country as a measure of tourism demand, and in some instances, the number of overnight stays or the tourist expenditures, both in aggregate and per capita forms, as presented by Song & Li (2008) in an overview of research in tourism demand modeling. Song *et al.* (2010) examine the most appropriate measures of tourism demand and point out that tourist expenditures will mostly be influenced by the current price levels. As presented in a study by Claveria *et al.* (2015), tourist arrivals show to be a more accurate measure in forecasting tourism demand, compared to using the number of overnight stays. In this thesis, both arrivals and overnights are included as measures of tourism demand to ensure model robustness and explore model performance and tourism determinants across different measures. One of the challenges of forecasting tourism demand, even with the availability of numerous studies and

related literature, is that there is no general model or rules that can be followed. The thesis will therefore rely on a combination of literature-backed results and its own analysis.

The selection of explanatory variables in this thesis is guided by a combination of findings from relevant literature. Claveria *et al.* (2015) rely exclusively on autoregressive terms, such as past values of tourist arrivals or overnights, to capture temporal dependencies and seasonality. In contrast, Chang-Jui *et al.* (2011) and Alamsyah & Friscintia (2019) emphasize the importance of incorporating macroeconomic indicators that influence tourist consumption, such as the unemployment rate, exchange rate, and consumer price index (CPI). Onder (2017) includes Google Trends data due to their potential to improve tourism demand forecasting. In addition, binary indicators for Schengen Area and Eurozone membership are introduced based on the results from Tkalec (2024) that point to a positive relationship between Croatia's EU integration and growth in tourism demand expressed in overnight stays. By combining autoregressive components, macroeconomic indicators, signals on search interests, and structural policy indicators, the models in the thesis aim to provide a comprehensive representation of the factors that influence tourism demand in Croatia.

2.3 Challenges

Despite the growing volume of existing literature on tourism demand forecasting using artificial neural networks, there are still some limitations when it comes to generating highly accurate models that could be easily generalized for other territories. As pointed out by Song *et al.* (2010) and Shafiullah *et al.* (2019) findings from such analysis should be handled with caution if chosen to apply to different regions, as there is not exactly a one-size-fits-all model and findings should be revised and targeted for specific regions. A common recommendation mentioned in several related papers, including Song *et al.* (2010), Witt & Witt (1995) and Vu & Turner (2005) mentions that future researchers should focus on including disaggregated data and using higher data frequency by employing monthly data instead of annual, which this thesis aims to overcome, along with other gaps in literature through the inclusion of data disaggregated in terms of country of origin of tourists, provided on a monthly basis.

Chapter 3

Methodology

This chapter presents the methodological framework adopted in this thesis, including the design and implementation of artificial neural networks (ANNs) for tourism demand forecasting. The following sections introduce the chosen methodological approach and all the steps following it.

3.1 Artificial Neural Networks

Artificial Neural Networks (ANNs) represent the core mechanism that is behind deep learning. They are modeled according to the brain's structure and cognitive processes that allows them to learn from data and detect complex patterns, Haykin (1994), Bishop (1995), Drew & et al. (2000), Schmidhuber (2015). In contrast to classical econometric methods, ANNs perform notably well when dealing with non-linear data patterns, which can be present in tourism demand, particularly when it depends on a wide range of interconnected determinants. ANNs are made up of multiple interconnected units called the neurons, which are organized into different layers. As explained by Saif (2023), these units are processing elements that can take input data and run mathematical computations based on defined activation functions, after which they pass the results forward through the network. According to literature, including Huang *et al.* (2024) and Saif (2023), being able to adjust their internal parameters such as weights and biases during training allows ANNs to both learn from data and refine their predictions over time. In the thesis, this optimization is done using the Adam optimization algorithm. The thesis employs a multilayer feedforward perceptron (MLP) type of ANN, that is based on a Feed Forward Neural Network (FFNN) architecture. Unlike Recurrent Neural Networks that enable

cyclical connections inside the network structure and feed the outputs back into the first layer, the MLP moves input data only in one direction, through the hidden layers and towards the output layer. Once this forward-pass is completed, the errors are back-propagated through the network and the weights and biases are readjusted to minimize the errors. The model architecture used in this research includes three main layers, resembling a structure as illustrated in Figures 3.1 and 3.2. The input layer includes historical information on all of the selected variables that are deemed relevant for the thesis analysis. Detailed information on the the input layer and the selected variables in this thesis can be found in Chapter 4 that explains in detail the steps involved in data collection and preprocessing. The information from the input layer is transferred through hidden layers to the output layer which generates the final result. These layers process the input data and extract meaningful patterns. Each neuron in a hidden layer applies a mathematical transformation to the inputs it receives. The exact number of hidden layers and neurons can be determined experimentally to optimize model performance. The output and the inputs have a relationship that can be presented with the following expression, also presented in works of Khashei & Bijari (2010); Teixeira & Fernandes (2014); Zhang & Qi (2005):

$$Y_k = b_k + \sum_{j=1}^n w_j f \left(\sum_{i=1}^m W_{ji} X_i + B_{0j} \right) \quad (3.1)$$

where Y_k ($k = 1, 2, 3, \dots, p$) is the output variable, B_{0j} and b_0 are meant to represent the biases associated with the nodes in hidden and output layers, W_{ji} ($i = 0, 1, 2, 3, \dots, m; j = 1, 2, 3, \dots, n$) and w_j ($j = 1, 2, 3, \dots, n$) are weights of connections between the nodes. W_{ji} is intended to connect input layer to the hidden layers and w_j to connect the hidden and output layers. m represents the number of nodes in the input layer and n refers to the number of nodes within the hidden layers. f is the chosen activation function. Activation Functions enable the network to learn and approximate relationships between input and output layers. With the aim of forecasting tourism demand, characterized by strong seasonality and nonlinear patterns, the hyperbolic tangent ($\tanh$) function is employed as the activation function within the hidden layers of the network. The $\tanh$ function maintains symmetry around zero and introduces non-linearity, which can lead to improved convergence during training.

$$\tanh(x) = \frac{e^x - e^{-x}}{e^x + e^{-x}} \quad (3.2)$$

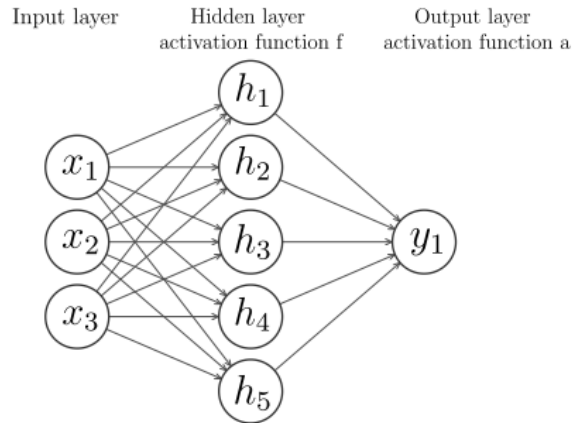
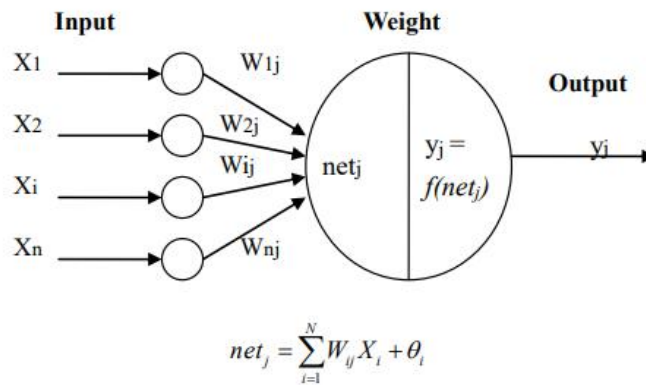


Figure 3.1: General Architecture of ANN

Figure 3.2: An Artificial Neuron (Chang-Jui *et al.* (2011))

In this thesis, the combined inputs were passed through two hidden layers that consisted of 64 and 32 neurons, using a tanh activation function to introduce non-linearity and generate an output layer of a single value that represents the next month's tourism forecast for Croatia. Traditional forecasting methods often struggle in accurately capturing the complex interactions between seasonality, external shocks and policy-driven shifts in demand. This thesis therefore focuses on an Artificial Neural Network (ANN) based forecasting approach, which is expected to offer more flexibility compared to conventional linear time series models. The ANN is capable of handling large datasets, modeling long-term dependencies in seasonal tourism trends and adapting to sudden disruptions such as the COVID-19 pandemic, EU and Schengen Area

accession or Euro adoption. By incorporating macroeconomic indicators like inflation, unemployment rates and exchange rates, the model is better equipped to generate accurate forecasts across different economic conditions.

3.2 Model Configuration

To ensure the ANN model is optimized for accuracy and reliability, this thesis carefully outlines its structure and configuration. The aim is to avoid overfitting, which may happen when the model performs perfectly on the training subset, but very poorly on the testing one, due to memorizing the training data, rather than learning from it and its underlying patterns, as supported by Feng *et al.* (2006) Wang *et al.* (2009).

3.2.1 Input Features

Each of the tested models incorporates both numerical and categorical inputs considered relevant for forecasting tourism demand. The numerical features include a range of lagged values of the target variables to capture temporal dependencies and seasonality and month indicators to reflect the recurring patterns in tourism flows. They also include macroeconomic indicators, such as unemployment rate, exchange rate, consumer price index.. All numeric features are standardized prior to modeling by applying Z-score standardization, which transforms each feature X in a way that gives it a mean of 0 and a standard deviation of 1, as shown in Equation 3.3

$$Z = \frac{X - \mu}{\sigma} \quad (3.3)$$

where X is the main input value, μ is its mean and σ represents the standard deviation. This ensures comparability across features and contributes to more stable model training.

Categorical inputs in machine learning can commonly be handled either through one-hot encoding or label encoding Hancock & Khoshgoftaar (2020). In this thesis, one-hot encoding was used for the variable indicating months, as there are only twelve possible values and this approach leads to a manageable increase in input dimensionality. The categorical variable indicating country of origin was handled with a two-step encoding procedure to avoid adding additional binary features, which would result from one-hot encoding more

than 20 countries. First, each country was assigned a unique integer label and then the label-encoded country input was passed through an embedding layer within the neural network, allowing the model to learn country-specific effects in a computationally efficient manner. Effective encoding of categorical data can be essential in predictive modeling. By transforming variables such as country of origin into a format that can be passed into the model, the neural network can capture group-specific effects and relationships, which ultimately enhances forecasting accuracy.

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3.2.2 Hyperparameters and Data Splits

The numeric input and the country embedding were concatenated and passed through two hidden layers with 64 and 32 units, using the hyperbolic tangent (tanh) activation function and the Mean Squared Error loss function as an evaluation metric. We tested the model with a Rectified Linear Unit (ReLU) activation function, but both had similar forecasting performances and the usage of tanh is supported by existing literature in this context Tuncsiper (2023); Alamsyah & Friscintia (2019). The output layer is a single neuron with linear activation that produces a continuous value, that is, the forecast for next-month arrivals or overnights. The thesis experimentally determined the final ANN configuration, including the number of hidden layers, neurons in them, learning rate, dropout rate, batch sizes and number of iterations.

The learning rate is in charge of controlling how quickly the model updates its weights during the training phase. It is important to find a balance that will avoid slow convergence, but simultaneously avoid using a very high learning rate that might cause overfitting. The thesis tested models for both target variables with learning rates of 0.001, 0.002, 0.003 and 0.005 and compared their evaluation metrics. Although the out-of-sample metrics presented in Table 3.1 point to slight improvements with higher learning rates, there are signs of overfitting and less stable convergence at learning rates above 0.001, as visible in Figure B.1. To keep the performance of the models comparable between each other and allow for robustness checks, the thesis will continue to estimate the models using a learning rate of 0.001.

Table 3.1: Performance metrics for different learning rates

LR	MSE	MAE	MAPE (%)	R ²	Target
0.001	0.4001	0.4916	5.71	0.8974	Arrivals
0.002	0.3585	0.4613	5.40	0.9081	Arrivals
0.003	0.3749	0.4638	5.47	0.9039	Arrivals
0.005	0.3587	0.4458	5.35	0.8980	Arrivals
0.001	0.3451	0.4549	4.52	0.9234	Overnights
0.002	0.3547	0.4652	4.57	0.9212	Overnights
0.003	0.2710	0.3909	3.98	0.9398	Overnights
0.005	0.3233	0.4257	4.25	0.9316	Overnights

Dropout helps avoid overfitting and helps the network generalize patterns on unseen data by randomly dropping a subset of neurons during the training phase Claveria *et al.* (2015). With regards to the size of the dataset and the common approach, a dropout rate of 20% is employed in each layer. The batch size determines the size of samples during training that are processed before model's weights and other internal parameters are updated. The number of epochs also needs to be defined as it represents the number of times the model finishes passes through the training dataset, as explained by Brownlee (2022). Devarakonda *et al.* (2018) point out that choosing a batch size can be a trade off between computational efficiency and cost. Following the existing empirical evidences provided in numerous literature and the compatibility with the Stochastic Gradient Descent (SDG) optimizers in tourism demand forecasting, this thesis uses adaptive moment estimation (Adam) optimization algorithm. It is a type of SDG optimizer that is considered efficient, fast converging and robust to the choice of hyperparameters, as it computes adaptive learning rates

for each parameter using first and second moments of the gradient which determine how much parameters should be adjusted to reduce the errors (Yi *et al.* (2020)). Figure 3.3 shows the loss curves for training and validation for both the arrivals and overnights models. The rapid convergence and the small gap between training and validation loss indicate stable training without overfitting.

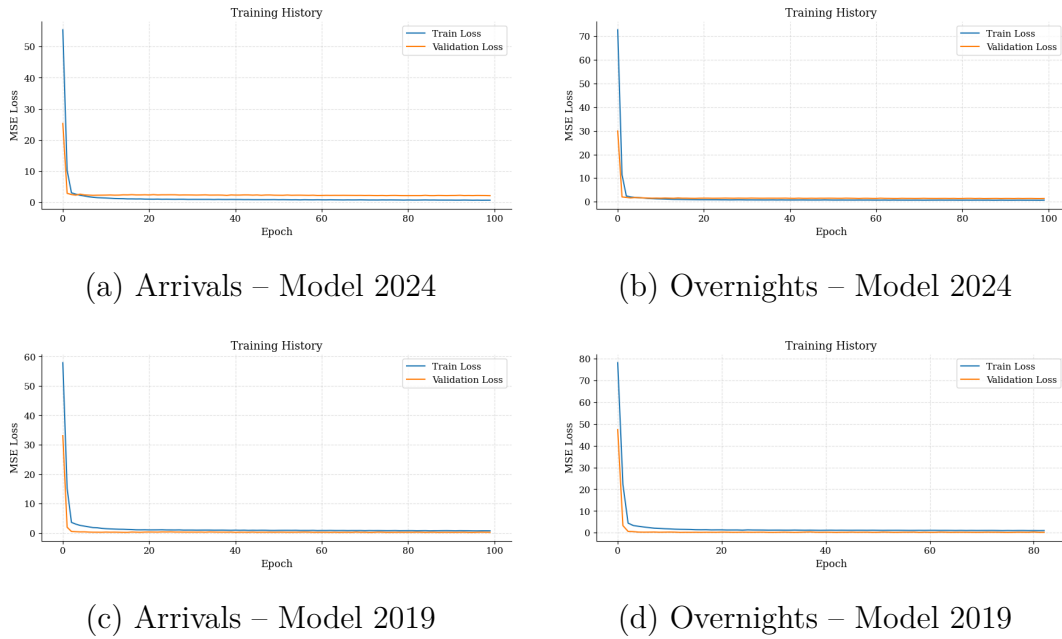


Figure 3.3: Training loss (MSE) across estimated ANN models

To ensure an unbiased evaluation of forecasting performance and minimize the risk of overfitting, the dataset was divided into three subsets: training, validation and test sets. This approach allows the models to train on one subset of data, fine-tune on a different portion of it and finally be evaluated on truly unseen observations.

In defining the training, validation and test splits, particular attention was paid to the chronological splitting of data to prevent data leakage and to the major structural shocks affecting tourism demand, most notably the COVID-19 pandemic. By dividing the dataset into full calendar years and isolating post-pandemic recovery years within the validation and test sets, the splitting approach ensures that the model is trained exclusively on stable, pre-pandemic data in certain variants and evaluated on periods marked by significant disruption in others. This design allows for an unbiased assessment of model performance across both routine and shock-affected periods and helps ensure that findings are robust to shifts in underlying demand dynamics. The approach is in line with recent literature emphasizing the importance of split representation

in the presence of global shocks Inoue *et al.* (2017). The specific date ranges for the splits are presented in Tables 3.2 and 3.3.

Table 3.2: Chronological data splits for models with data until 2024

Set	Start Date	End Date	Duration (months)
Train	2001-01	2016-12	192
Validation	2017-01	2020-12	48
Test	2021-01	2024-12	48

Table 3.3: Chronological data splits for models with data until 2019

Set	Start Date	End Date	Duration (months)
Train	2001-01	2014-12	168
Validation	2015-01	2017-12	36
Test	2018-01	2019-12	24

3.3 Benchmark Model and Model Evaluation

This thesis uses a simple autoregressive AR(12) model as a benchmark for evaluating the performance of more complex forecasting techniques. This model represents a traditional univariate time series approach that relies on the values from the previous twelve months to make forecasts and is applied separately to each target variable, making it suitable for datasets with strong seasonal cycles. In the context of tourism demand, where annual seasonality and temporal persistence are common, such a model offers a straightforward way to capture recurring patterns without relying on additional explanatory variables. While more flexible models may offer advantages in capturing nonlinearities or incorporating external factors, the AR(12) model remains a standard point of reference in applied forecasting studies due to its transparency, ease of estimation and consistent performance in well-behaved time series Song & Li (2008); Athanasopoulos *et al.* (2011). The AR (12) model was estimated using the log-transformed monthly target variables' series. In cases where the series did not satisfy stationarity assumptions, first differencing was applied.

The performance of the employed models in this thesis is evaluated using four typical evaluation metrics: Mean Squared Error (MSE), Mean Absolute Error (MAE), Mean Absolute Percentage Error (MAPE) and the coefficient of determination (R^2). They provide complementary perspectives on model accuracy by capturing the prediction errors, their proportional size relative

to observed values and the proportion of variance explained. They are used consistently throughout the analysis to evaluate and compare the forecasting performance of the ANN and benchmark models.

The Mean Absolute Error (MAE) represents the average magnitude of errors between actual and predicted values. In doing so, it does not considering their directions. As mentioned by Goodfellow *et al.* (2016); Terven *et al.* (2024), MAE can be slightly less sensitive to outliers, as opposed to MSE, as it weights all individual differences equally. This makes MAE a robust and intuitive measure for assessing overall model accuracy. The formula is:

$$\text{MAE} = \frac{1}{n} \sum_{i=1}^n |y_i - \hat{y}_i| \quad (3.4)$$

where y_i are actual values, $\hat{y}_i$ are the predicted values and n is the number of observations. Lower MAE values indicate higher predictive accuracy. Compared to MSE, MAE is more robust to large errors but is not differentiable at zero, which can affect certain optimization algorithms.

The Mean Absolute Percentage Error (MAPE) represents the average percentage error between actual and predicted values, making it particularly useful for comparing forecasting performance across datasets with different scales of the target variable. As explained by Terven *et al.* (2024), its scale independence and intuitive interpretation in percentage terms are key advantages. MAPE is computed as:

$$\text{MAPE} = \frac{1}{n} \sum_{i=1}^n \left| \frac{y_i - \hat{y}_i}{y_i} \right| \times 100 \quad (3.5)$$

where y_i are the actual values, $\hat{y}_i$ the predicted ones and n is the number of observations. Lower MAPE values indicate more accurate predictions.

The Root Mean Squared Error (RMSE) aims to show the average magnitude of prediction errors. It does so using the square root of MSE, that is the average squared differences between the actual and predicted values:

$$\text{RMSE} = \sqrt{\frac{1}{n} \sum_{i=1}^n (y_i - \hat{y}_i)^2} \quad (3.6)$$

where y_i represents the actual values, $\hat{y}_i$ the predicted values and n the number of observations. RMSE is expressed in the same units as the target variable, making it highly interpretable in practical applications. It puts more penalties on larger errors than smaller ones, which can be advantageous when large

deviations are particularly undesirable. However, it is sensitive to outliers and does not distinguish between types of error Terven *et al.* (2024).

The coefficient of determination, denoted as R^2 , calculates the portion of variance in data that can be explained by model's predictions. According to Terven *et al.* (2024), it aims to show the explanatory power of the model, with values in a range from 0 to 1, from no explanatory power to a perfect fit. Negative values may occur if the model's predictions are worse than just predicting the data means. The formula is:

$$R^2 = 1 - \frac{\sum_{i=1}^n (y_i - \hat{y}_i)^2}{\sum_{i=1}^n (y_i - \bar{y})^2} \quad (3.7)$$

where y_i denotes the actual values, $\bar{y}$ their mean and $\hat{y}_i$ are the predicted values. Higher R^2 values indicate better model fit. To evaluate which input variables are the most influential on the model's predictions, we employ permutation feature importance, which shows how much the model's performance worsens when the values of one feature are randomly shuffled. The results are visually presented using bar plots. Breaking the relationship between that feature and the target variable allows us to see how important it was for the model's prediction accuracy. A larger drop in accuracy indicates a more important feature. For seasonal effects, all monthly dummy variables are shuffled together to reflect their joint impact. This helps identify which variables contribute most to forecasting tourism demand in Croatia.

To ensure the model is robust and findings are reliable, this thesis incorporates several additional validation steps. In addition to forecasting tourism demand using arrivals as the main target variable, model performance is also evaluated using overnights as an alternative measure. This comparison helps confirm that the results are consistent across different definitions of demand. The modeling approach relies on a chronological train/validation/test split, with early stopping that aims to prevent overfitting by interrupting training once validation errors stop improving. This procedure ensures that the model's predictive accuracy generalizes well to new, unseen data. All preprocessing and modeling steps were carried out in a reproducible manner, using fixed random seeds and standardized data preparation.

Chapter 4

Data

4.1 Overview of the Dataset

The dataset employed in the empirical section of the thesis includes monthly observations from 2000 until end of 2024 across Croatia and 23 countries of origin. Tourism demand is measured through the number of monthly tourist arrivals and overnights stays in Croatia. For additional clarity and insights, the data is grouped by countries of origin of the tourists. Both the tourist arrivals and overnight stays are sourced from the Croatian Bureau of Statistics that collects data registered in the Croatia's official administrative system, eVisitor. The list of countries of origin, for which explanatory variables are collected is presented in Table 4.1.

The twenty three countries of origin have been selected according to their contributions to Croatia's tourism to ensure a diverse and representative sample of Croatia's tourism demand and improve model robustness and applicability. The list includes both the countries with high volumes of arrivals and overnights as well as those with smaller or moderate numbers, to give the models opportunity to capture varying economic conditions, market dynamics and travel behaviors, rather than overfitting to patterns that are specific to only high-volume markets.

Country	Country
Austria	Belgium
Bosnia and Herzegovina	Canada
Czech Republic	Denmark
Finland	France
Germany	Hungary
Ireland	Italy
Netherlands	Norway
Poland	Romania
Slovakia	Slovenia
Spain	Sweden
Switzerland	United Kingdom
United States of America	

Table 4.1: List of countries of origin included in the analysis

The final dataset includes several macroeconomic indicators for each country of origin, along with their respective Google Trends data related to tourism searches for Croatia. To prevent data leakage and ensure temporal consistency, most variables were used in their lagged forms to reflect only the information that would have been available at the time of prediction and in the case of arrivals and overnights, to capture autocorrelation patterns and improve model’s ability to learn from past data.

The input variables included in the model were collected from a combination of international and Croatian public databases to ensure the credibility and transparency of the dataset. Macroeconomic indicators for each country of origin were primarily sourced from the World Bank and Eurostat, the respective national statistical bureaus and central banks.

Monthly unemployment rates were obtained mainly from the OECD Data Explorer and supplemented by UNECE for countries not covered by OECD. Exchange rate data, defined as national currency units (NCU) per euro, was retrieved from UNECE. Consumer Price Index (CPI) values (base year 2010=100) were also sourced from UNECE, while industrial production indices were initially considered but excluded from the final model due to a high portion of missing values and insignificant contribution.

Information on each country’s Schengen Area membership and the year of accession was collected from the European Commission, while the timeline of euro adoption was sourced from the European Central Bank. These policy-related variables were encoded as binary indicators, taking the value of 1 for years where it is true.

In addition to macroeconomic and policy variables, the dataset includes Google Trends data for the search interest in Croatia from each origin country, using tourism-related keywords. This data spans from 2004 to the end of 2024 and is meant to serve as an indicator for country-specific demand signals related to Croatian tourism. Since Google Trends data is not available for the years 2000–2003, these missing values were imputed with zero values. Table 4.2 shows the descriptive statistics of the key variables prior to any log transformation, lagging, or feature engineering. Tourist arrivals vary from zero to over one million, while overnight stays reach up to eight million, indicating substantial differences in travel volume. These large differences, along with high standard deviations, reflect strong seasonal patterns and long-term changes in tourism activity. Economic indicators, such as the unemployment rate and exchange rate also show a lot of variability, highlighting the changing economic environment over the years across the different countries. The inclusion of EU, Schengen and Euro membership as binary indicators aims to help in capturing the effects of important policy changes on tourism trends.

Statistic	Arrivals	Overnights	Unemp. Rate	Exch. Rate	CPI	Schengen	Euro	EU Member	Google Trends
Count	6,898	6,898	6,898	6,898	6,898	6,898	6,898	6,898	6,898
Mean	35,403.7	198,116.4	7.86	16.44	103.89	0.6606	0.451	0.7357	19.13
Std	78,306.9	547,866.8	4.85	61.21	18.14	0.4735	0.498	0.4409	19.10
Min	0	0	1.6	0.589	27.4	0	0	0	0
25%	1,305.5	3,667.5	4.8	1.0	91.93	0	0	0	2.0
50%	6,301.5	20,108.0	6.8	1.0	103.7	1	0	1	15.0
75%	28,417.5	110,476.8	8.8	4.38	111.5	1	1	1	31.0
Max	1,107,217	8,550,147	31.13	418.31	186.9	1	1	1	100.0

Table 4.2: Summary statistics of selected input variables

4.2 Tourism Demand: Arrivals and Overnights

This section explores the dynamics of tourism demand in Croatia, with a focus on monthly tourist arrivals and overnight stays across different countries of origin. Understanding these long-term trends is necessary for identifying seasonal patterns, shocks and the impact of policy changes that can help shape inputs for the neural network model.

Presented in Figure 4.1 is the total number of tourist arrivals and overnights in Croatia aggregated over the full dataset period by country of origin. Germany stands out with a significantly higher volume of both arrivals and overnights. There are considerable differences between the number of arrivals and overnights across countries, suggesting varying patterns of stay duration and travel behavior.

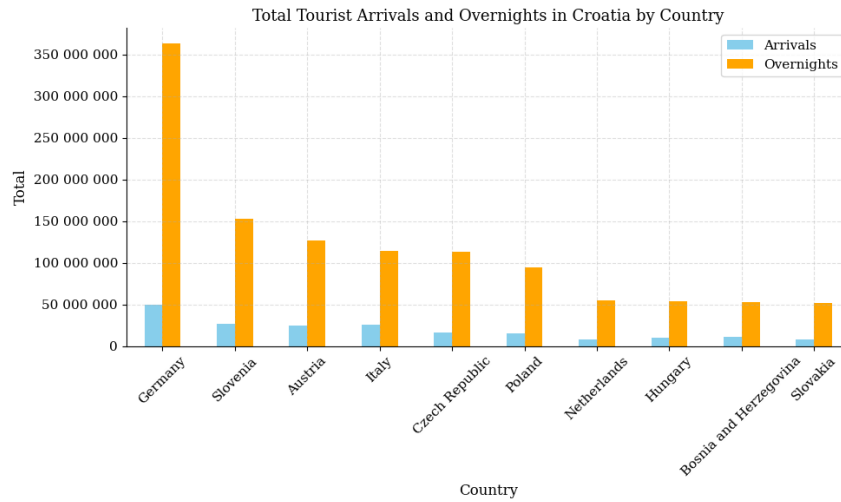
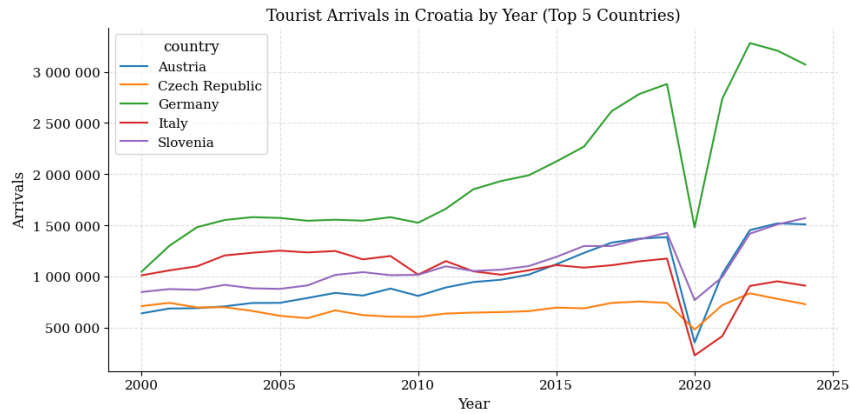
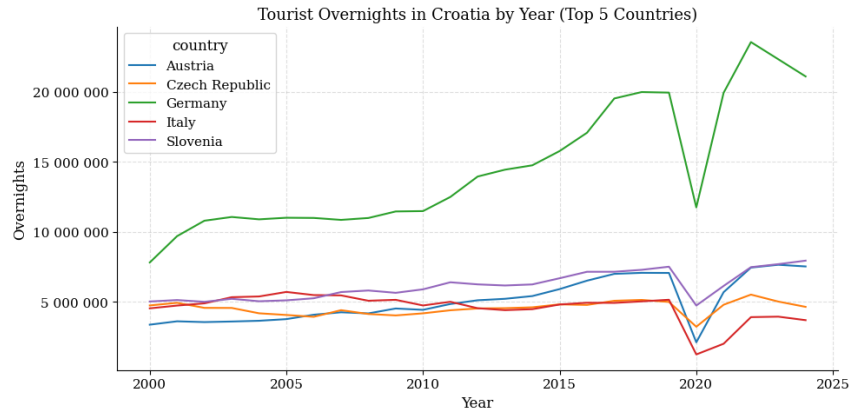


Figure 4.1: Total Tourist Arrivals and Overnights by Country

Figure 4.2 provides additional insight into the tourism demand of Croatia over the past 25 years and its behaviors across five leading countries of origin, including Germany, Slovenia, Italy, Austria and the Czech Republic. All five countries show an overall increasing trend in tourism demand over the observed period, with a sharp decline in 2020 due to the COVID-19 pandemic, followed by a gradual recovery in the subsequent years. Some countries, particularly Germany, Slovenia and Austria even surpassed their pre-pandemic levels, suggesting a strong rebound in travel demand. Germany consistently and significantly outperforms other countries in its tourism demand for Croatia and has the most pronounced post-pandemic recovery, with the tourism volumes exceeding their pre-pandemic levels. This highlights Germany's role as Croatia's most important tourism market and shows its resilience even in the face of external shocks.



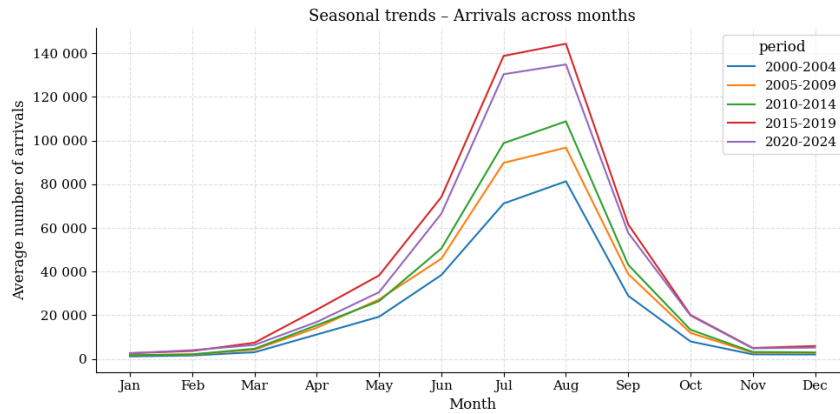
(a) Tourist Arrivals in Croatia by Year (Top 5 Countries)



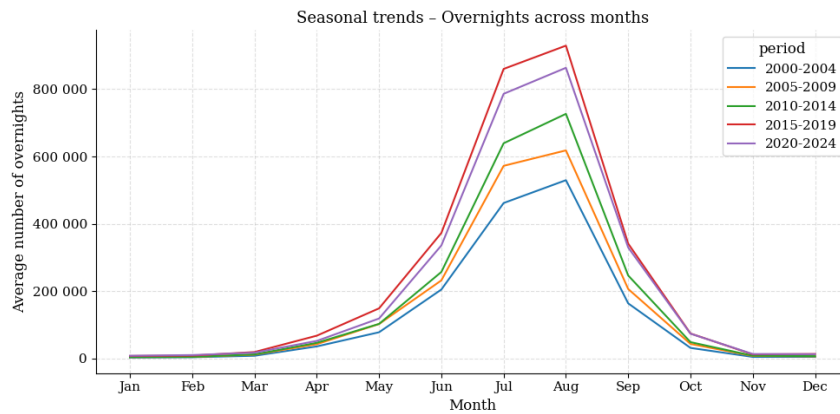
(b) Tourist Overnights in Croatia by Year (Top 5 Countries)

Figure 4.2: Annual trends of tourist arrivals and overnights in Croatia from the leading countries of origin of tourists (2000—2024).

Tourism demand in Croatia has strong seasonal fluctuations, with the peak tourism months concentrated between June and August, making it a highly seasonal, summer destination. This seasonality is a key consideration in forecasting demand, as it strongly influences monthly demand patterns. Figure 4.3 shows the monthly average arrivals and overnights across five-year intervals from 2000 to 2024. While the general seasonal trend remains stable, there is a visible upward shift in both arrivals and overnights over time, reflecting Croatia's growing popularity as a tourist destination. In more recent periods, a gradual increase in arrivals and overnights in non-summer months is also observable, which may signal a gradual extension of the peak season, potentially influenced by warmer weather conditions outside of the traditional summer months.



(a) Seasonal trends in Croatia's tourism arrivals across five-year subsets



(b) Seasonal trends in Croatia's tourism overnights across five-year subsets

Figure 4.3: Monthly seasonal patterns of tourism arrivals and overnights in Croatia across five-year periods

As shown in Figure 4.4, Croatia's monthly tourism demand has experienced a notable growth over the observed time span. The growth in monthly overnights is even steeper, indicating a gradual and significant rise in average duration of tourist stays in Croatia. This distinction may explain why forecasting overnights resulted in slightly better model performance than for arrivals, as it captures not only the number of tourists but also their length of stay. Notable shifts are visible around key moments, such as a clear decline in 2020 due to the COVID-19 pandemic and travel restrictions caused by it, followed by a strong recovery in the subsequent years. Vertical dashed lines mark Croatia's EU accession in 2013 and Eurozone and Schengen Area entry in 2023, mark-

ing its important policy changes, without implying a direct causal relationship with the observed trends.

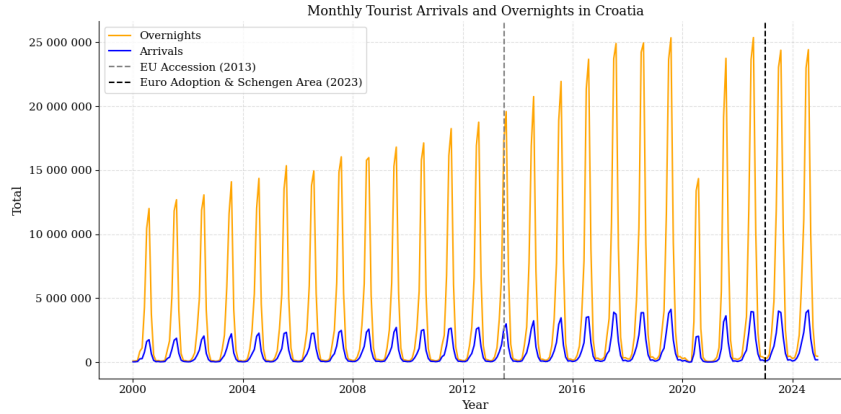


Figure 4.4: Total Monthly Arrivals and Overnights in Croatia.

To observe the long-term trend in tourism demand in Croatia, we applied a LOESS smoothing technique as part of the seasonal-trend decomposition. LOESS is a non-parametric method that fits locally weighted regressions across the time series, allowing for flexible extraction of nonlinear trends Cleveland *et al.* (1990); Jacoby (2000). Although existing literature, including Zhang & Qi (2005); Kihoro *et al.* (2004), argues that there is no need for seasonal adjustment prior to applying artificial neural networks, as these models can directly capture both seasonal and trend components, LOESS was applied here solely for visualization purposes and presented in Figure 4.5. This approach helps highlight how the overall trend evolved over time, especially around structural events such as Croatia’s Schengen accession, COVID-19 pandemic and the Euro adoption without requiring explicit seasonal component elimination.

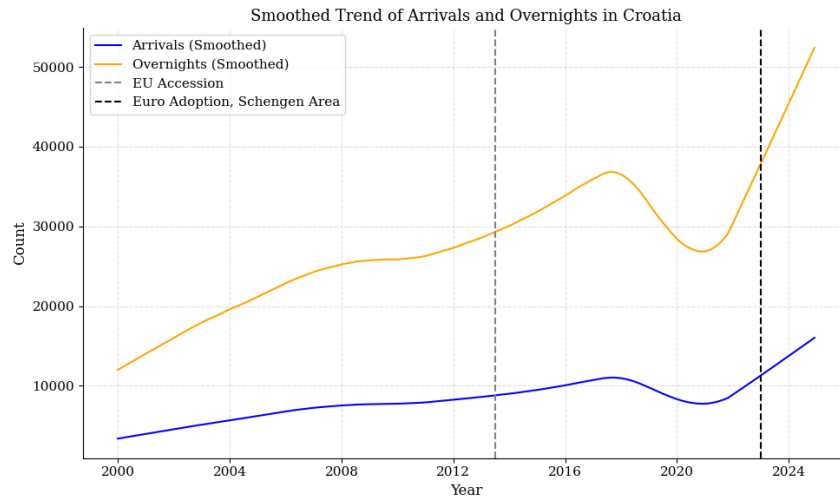


Figure 4.5: Tourism demand trend using LOESS smoothing method

Multiplicative seasonal-trend decomposition using LOESS in Figure 4.6 was applied to total arrivals to separate the long-term trend, seasonal fluctuations and residual noise. The smoothing parameter was set to 0.25, meaning that for each time point, LOESS fits a local regression using the nearest 25% of observations. This revealed increasing seasonal intensity over time and highlighted disruptions such as the COVID-19 shock. However, for our further analysis we do not remove the seasonal components, as we expect that the model trained with the seasonal effect will be more accurate at predicting tourism demand and will provide robust forecasting performance, as numerous literature already suggests Crone & Dhawan (2007); Nelson *et al.* (1999).

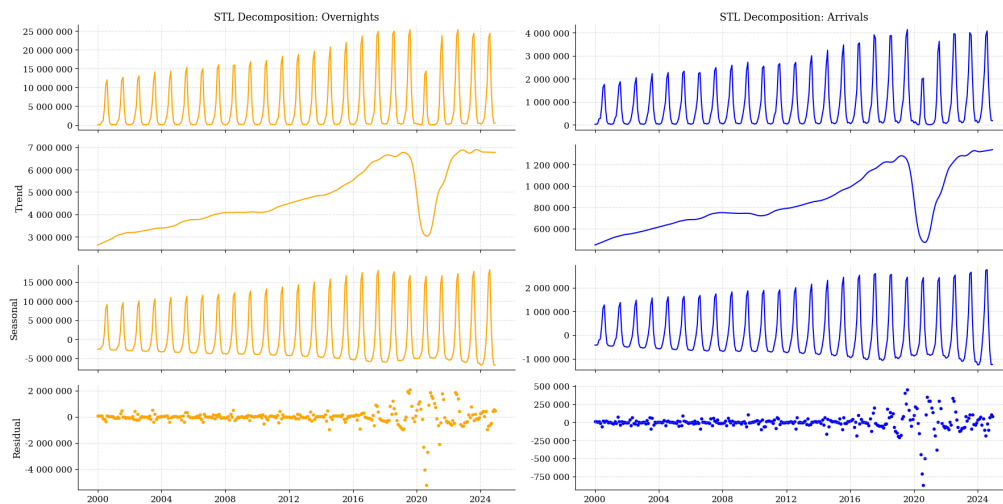


Figure 4.6: Seasonal-Trend Decomposition Using LOESS

To examine temporal dependencies in the data, we plot the autocorrelation

function (ACF) for monthly overnights and arrivals. As shown in Figure 4.7, both time series exhibit a strong positive autocorrelation at lag 1, indicating short-term persistence and a clear seasonal pattern with significant positive autocorrelation observed at lag 12, representative of the annual seasonality. The ACF plots also show significant negative autocorrelation at lags 3 and 6, suggesting the opposite directions across quarterly and semi-annual intervals, which reflects Croatia's tourist off-season. Although the correlation at these lags is negative, their statistical significance implies that these time points still contain useful predictive information. Therefore, the thesis includes the 3- and 6- month lagged variables in the model as well, as they may improve model's ability to capture regular cyclical changes in travel demand. The patterns observed in the ACF plots support the inclusion of lags 1, 3, 6 and 12 as input features in the forecasting model, capturing both recent trends and seasonal fluctuations in tourism activity.

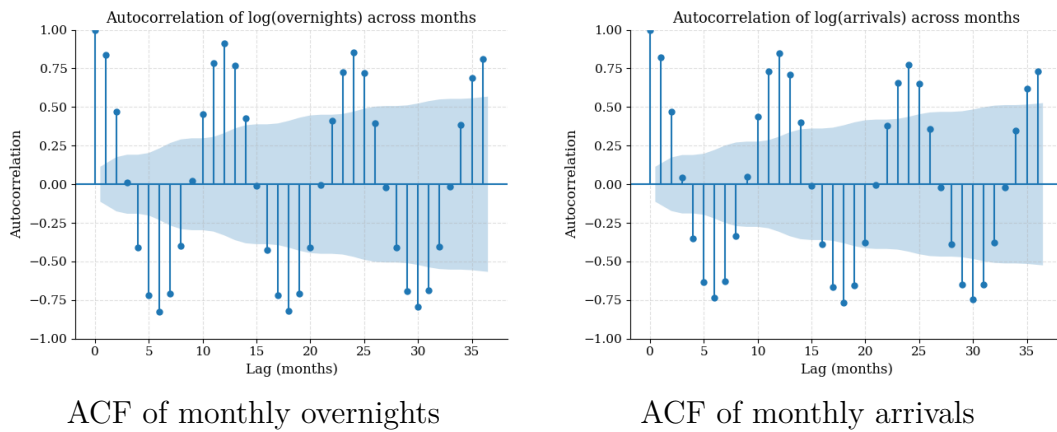


Figure 4.7: Autocorrelation function (ACF) plots for log-transformed monthly overnights and arrivals

4.3 Data Preparation and Cleaning

Before proceeding with the neural network model and forecasting the values, data was cleaned and preprocessed to ensure it was consistent and accurate. Variables were converted to appropriate numerical types. Columns such as tourist arrivals, overnight stays, unemployment rate, exchange rate and consumer price index were cast as floating-point numbers, while policy variables such as Schengen membership and euro adoption were encoded as binary integers. To capture fluctuations in public interest toward Croatia as a travel

destination, the thesis includes Google Trends data, which measures the relative frequency of specific keyword searches over time. For each origin country, monthly search intensity was retrieved for the keyword “Croatia”, focusing on tourism-related search behavior. This helps reflect demand-side signals that may not be captured by traditional macroeconomic indicators. As Google Trends data is only available from 2004 onwards, values for earlier years were imputed as zero. All other input variables were complete and required no imputation.

To normalize the distribution and reduce the impact of extreme values, a logarithmic transformation was applied to the number of tourist arrivals, overnight stays, CPI and exchange rates. After log-transformation, all of the input numeric features were standardized using a StandardScaler to ensure they have a mean of zero and variance of 1, to enable faster and more stable training of the neural network, so that all inputs contribute equally to the learning process. These steps were done with the aim to stabilize variance and improve convergence of the final model. To enable the model to learn temporal dependencies and autocorrelation patterns in tourism behavior, several lag features were created. These included 1-, 3-, 6- and 12-month lags of both arrivals and overnights to reflect the short-term, quarterly, semi-annual and annual patterns in travel demand. For Google Trends, 1- and 3-month lags were included under the assumption that travel-related online searches may occur one to a few months prior to the actual trip, potentially making them early indicators of future tourism demand. A 1-month lag of CPI and a 3-month lag of unemployment rate were included because these indicators are typically published with a time delay and less frequent intervals. Therefore, using their current values would not be available in real-time for monthly forecasting. All lag features were created within each country to maintain temporal integrity and prevent data leakage. However, the introduction of the lagged variables introduced some missing values at the beginning of each country’s time series, which was handled by restricting the dataset to start from January 2001. However, data from the year 2000 is still indirectly and partially reflected through the lag features. Dataset also includes calendar months which have been one-hot encoded to capture seasonal effects. Although this approach has some limitations, such as not capturing the true cyclical distance between months, the drawback is not critical in the context of this model, which aims to highlight seasonality and summer peaks, rather than modeling the proximity between all of the months. One-hot encoding months was chosen for its simplicity, efficiency and proven

effectiveness in neural network models, as successfully shown in prior studies such as Duan (2019).

The selection of input variables was supported by theoretical assumptions and prior empirical research, including Song & Li (2008); Song *et al.* (2010); Claveria *et al.* (2015); Chang-Jui *et al.* (2011), as well as the calculated Pearson correlation matrix and retroactively by the model's feature importance scores. These scores provided insight into the contribution of each variable on the model's predictions and have been supported in works of Sandri & Zuccolotto (2006); Sanchez-Pinto *et al.* (2018); Hapfelmeier & Ulm (2013); Genuer *et al.* (2010). Altogether, this supported the idea that a certain combination of indicators on macroeconomic conditions, important policy changes and online search trends can give insights into the tourism demand of the observed country and simultaneously affect it. Table 4.3 shows the correlation coefficients between the input features and the target variable calculated using the Pearson correlation matrix provided in Figure A.1.

Table 4.3: Correlation of model input features with target variables arrivals and overnights

Feature	Corr. with Arrivals	Corr. with Overnights
arrivals_lag_1	0.99	0.92
arrivals_lag_3	0.63	0.61
arrivals_lag_6	0.35	0.34
arrivals_lag_12	0.81	0.81
overnights_lag_1	0.91	0.94
overnights_lag_3	0.42	0.57
overnights_lag_6	0.28	0.32
overnights_lag_12	0.83	0.99
google_trends_lag_1	0.45	0.42
google_trends_lag_3	0.43	0.40
cpi_lag_1	0.15	0.12
unemployment_rate_lag_3	-0.01	-0.01
exchange_rate	-0.06	-0.06
eu_member	0.13	0.08
schengen_member	0.06	0.08
euro_adopted	0.13	0.13

As expected, recent lags show the strongest positive correlations, reflecting persistence in tourism flows. In contrast, the six-month lag exhibits a noticeably weaker correlation, which may be attributed to seasonality, as a half-year difference often corresponds to different travel patterns. Google Trends data

show moderate positive correlations with the target, supporting their inclusion in the model as leading indicators. Variables representing unemployment rates and exchange rates show weak negative correlations with the target variable, consistent with economic expectations that higher unemployment or a less favourable exchange rate may reduce the likelihood of travel.

Chapter 5

Results and Discussion

This section of the thesis shows the performance outcomes of five neural network models developed to forecast monthly tourism demand in Croatia. The chosen architectural structure of the feedforward neural network remains consistent throughout the different variations, while the models differ in terms of the target variables, time periods and input features. Some models are forecasting number of tourist arrivals and others number of tourist overnights. The time period varies between the models to observe its performance when it controls for the effects of the pandemic. These variations were introduced to evaluate the robustness and generalization of the forecasting approach under different conditions.

5.1 Comparing Model Performance

The final structure of the neural network has been chosen by experimental techniques and findings from existing literature on the similar topics. As defined in Chapter 3, the network combines numerical inputs with categorical embedding for countries of origin. Specifically, each model used two inputs: (1) a vector of macroeconomic and tourism-related numeric features processed via a fully connected layer, and (2) an embedding vector representing the origin country, that was learned from the categorical country variable and mapped to a dense representation space of 9-15 dimensions. The models were compiled using a consistent learning rate of 0.001 and the Adam optimization algorithm. The dataset was split chronologically within each country, with approximate 70% of the data used for training and remaining 30% for validation and testing. Early stopping was employed to prevent overfitting, monitoring validation loss with

a patience of 40 epochs. The training performances for both target variables are presented in Figure 3.3. Table 5.1 summarizes the performance metrics for the estimated ANN models and compares them to a benchmark autoregressive (AR) model calculated using the past 12 lags of target variables, serving as a baseline for evaluating ANN improvements.

Table 5.1: Performance metrics by target variable, time period, and model type for out-of-sample performance

Target	Period	Model	MSE	MAE	MAPE (%)	R^2
Arrivals	2019	ANN	0.3844	0.4703	5.22	0.8797
Overnights	2019	ANN	0.3737	0.4898	4.85	0.9092
Arrivals	2024	ANN	0.4001	0.4916	5.71	0.8974
Overnights	2024	ANN	0.3451	0.4549	4.52	0.9234
Arrivals	2019	AR	0.0275	0.1311	7.78	0.9542
Overnights	2019	AR	0.0405	0.1646	16.25	0.9539
Overnights	2024	AR	1.1755	0.6296	5.09	0.6856
Arrivals	2024	AR	2.0309	0.8312	8.24	0.4076

The performance metrics show that all neural network models achieved solid results in forecasting tourism demand. Based on the MAPE threshold proposed by Lewis (1982), where values below 10% are considered accurate, all models meet the standard for acceptable forecast accuracy.

Neural networks trained on the full dataset consistently outperformed those trained only on data up to 2019. This suggests that including the COVID-19 shock and the recovery period helps the model generalize better by learning from periods of disruption as well as normalization. For example, the model predicting overnights using the full dataset achieved a higher R^2 (0.9268) and lower MAPE (4.49%) than the same model with a 2019 cutoff.

Models using overnights as the target variable also performed slightly better than those using arrivals. The best-performing model overall was a neural network trained on overnights across the full period. Although Claveria *et al.* (2015) argue that arrivals are more accurate than overnights at predicting tourism demand in their example of Catalonia, the evidence from this thesis has the opposite finding. Overnights are shown to be a slightly more stable and informative indicator of tourism demand than arrivals, possibly due to their stronger correlation with trip duration and tourist spending. However, in the thesis we will continue to look at both of the best performing models and treat them as a mutual robustness measure.

In contrast, the AR(12) benchmark models performed noticeably worse, especially when trained on the full dataset. Their performance declined significantly in periods affected by external shocks, like the one caused by the COVID-19 pandemic. For instance, the AR model for arrivals in 2024 had a much higher error and a much lower R^2 (0.4076) than the neural network alternative. This suggests that AR(12) models might only be suitable for relatively stable periods and are not as capable of handling sudden shifts and complex, nonlinear patterns that appear in recent tourism data. However, even then they do not outperform neural network forecasts in predictive accuracy. A visual comparison of AR(12) forecasts and actual values for arrivals and overnights is shown in Figure 5.1. While the model captures the general seasonal structure, it tends to overestimate the actual values and is less adaptive to structural shifts. In conclusion, the most effective model structure was a feedforward neural network using country embeddings, macroeconomic lag features, and the full 2000–2024 dataset, with overnights as the target variable. This structure provided the a good trade-off between the predictive accuracy of the model and its complexity.

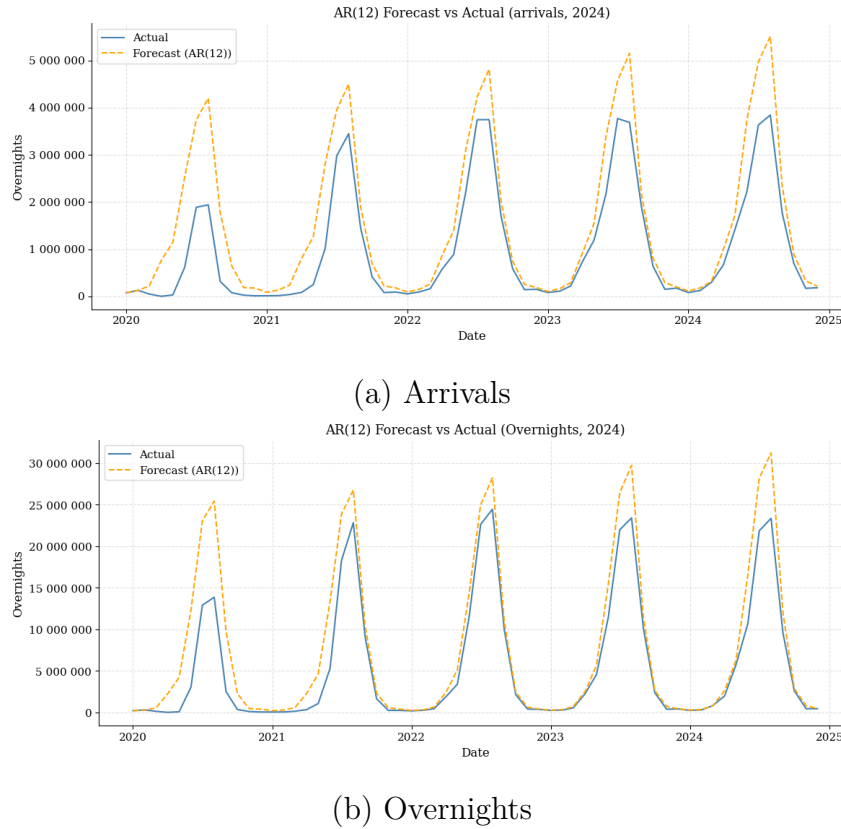
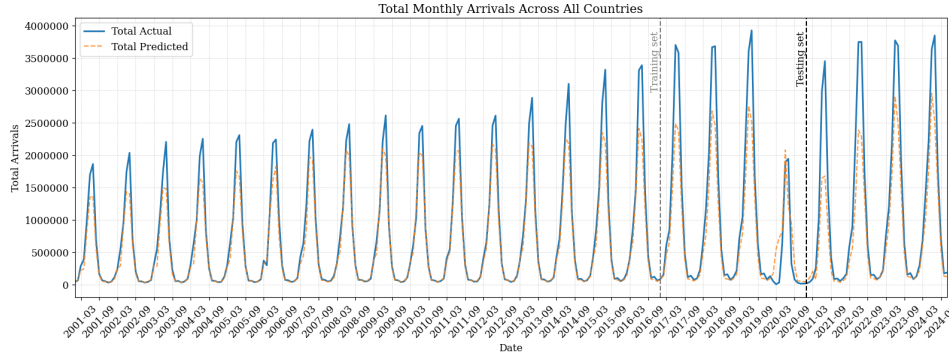
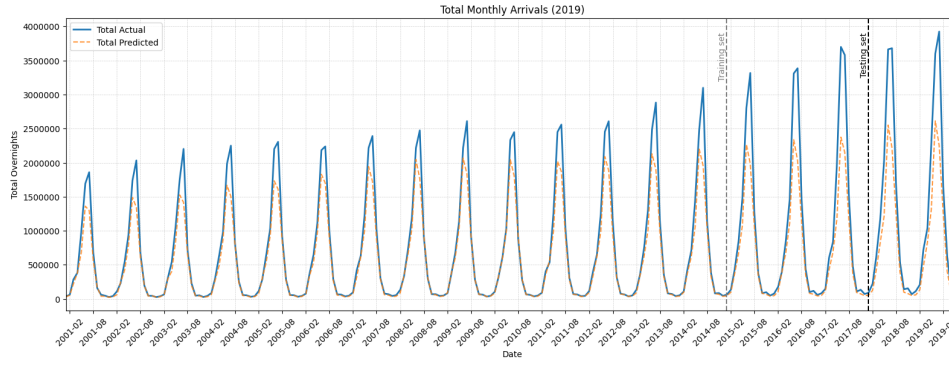


Figure 5.1: Predicted and actual monthly arrivals using benchmark model AR(12), 2020–2024

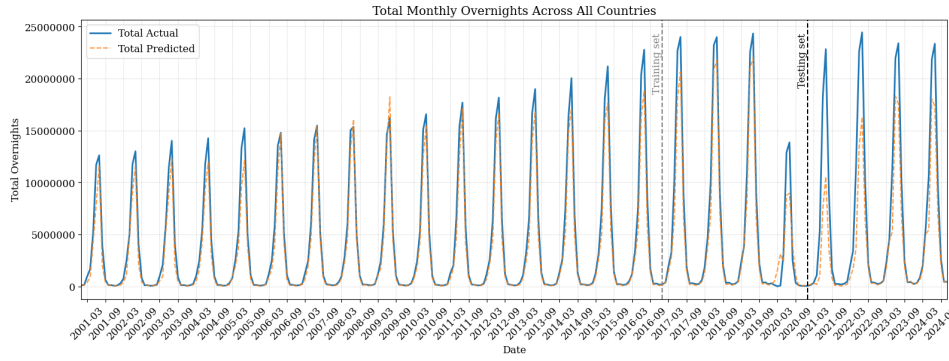
Both models, as seen in Figure 5.2 show a high accuracy in capturing the overall trend and seasonal patterns in the data. They continue to perform well even during the pandemic years (2020-2021), successfully predicting the sharp decline in tourism demand, even with the absence of any explicit pandemic-related parameters. This suggests that the models were able to predict the drop by responding to sudden changes in the explanatory variables, such as rising unemployment, falling exchange rates, declines in consumer prices and lower interest in travel-related search terms captured by Google Trends, which were among the key effects observed during the pandemic period Alon *et al.* (2021); Jawad & Naz (2023). This highlights the model's ability to learn non-linear relationships between macroeconomic conditions and tourism demand, without requiring manually engineered crisis indicators.



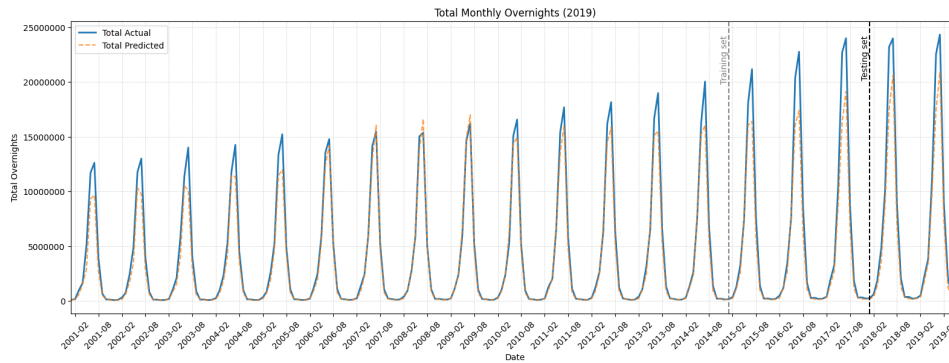
(a) Arrivals: ANN (2000-2024)



(c) Arrivals: ANN (2000-2019)



(c) Overnights: ANN (2000-2024)



(c) Overnights: ANN (2000-2019)

Figure 5.2: The actual values and the values obtained from the ANN for forecasted total arrivals and overnights (2000–2024)

5.2 Forecasting Performance by Country of Origin

To further assess if the model is able to generalize across different national origins, we evaluated forecasting performance at the country level. Figure 5.3 presents a panel of comparisons between the actual and predicted monthly overnights in Croatia over the entire time span from 2000 until 2024. The figure is included to provide context on tourism flows and its fluctuations, varying scales and trends across different countries. It also enables a visual inspection of both in-sample and out-of-sample model behavior. The model's performance on only the testing set is shown in Figure 5.4 to allow for a more granular inspection. Table 5.2 shows the performance metrics for each country of origin in the dataset. The results show that the model performs well for a lot of countries included in the analysis and tends to decently capture the strong seasonality and upwards trends in demand. Forecasts for countries with relatively stable or moderate tourism volumes, such as Canada, Finland, Bosnia and Herzegovina, closely follow the actual values, even during the uncertain post-COVID years and do relatively well at modeling tourism demand. For some origin countries, model underestimate the strength of the post-COVID recovery, particularly in 2022 and 2023, or predicts it with some delay. While it often captures the direction of change and seasonal shape of tourism demand, it tends to predict a slower or more modest rebound than what actually occurred. This limitation is especially noticeable in countries where demand surged quickly after travel restrictions were lifted, such as Norway or Ireland. Although the model includes macroeconomic and behavioral indicators that may signal recovery, it appears to struggle with fully translating those signals into higher volume peaks. This suggests that the network may be slightly restrictive under these circumstances, especially when no such dynamics were not present during the training set. The overall largest discrepancies are observed during periods of extreme shock, notably the pandemic, and in countries characterized by consistently above-average tourism demand. In some high-volume markets, such as Germany, Italy, USA, Poland and the Czech Republic, the model tends to underestimate the actual number of arrivals and overnights. Most likely, this can be attributed to both the use of log-transformed variables for forecasting which compress the scale of high values and reduce sensitivity to large variations for some of the countries with extremely high values and to the general tendency of neural networks to prioritize minimizing average error across the entire dataset. As a result, predictions for extreme values may be systematically biased toward

the overall mean Zhang *et al.* (2017).

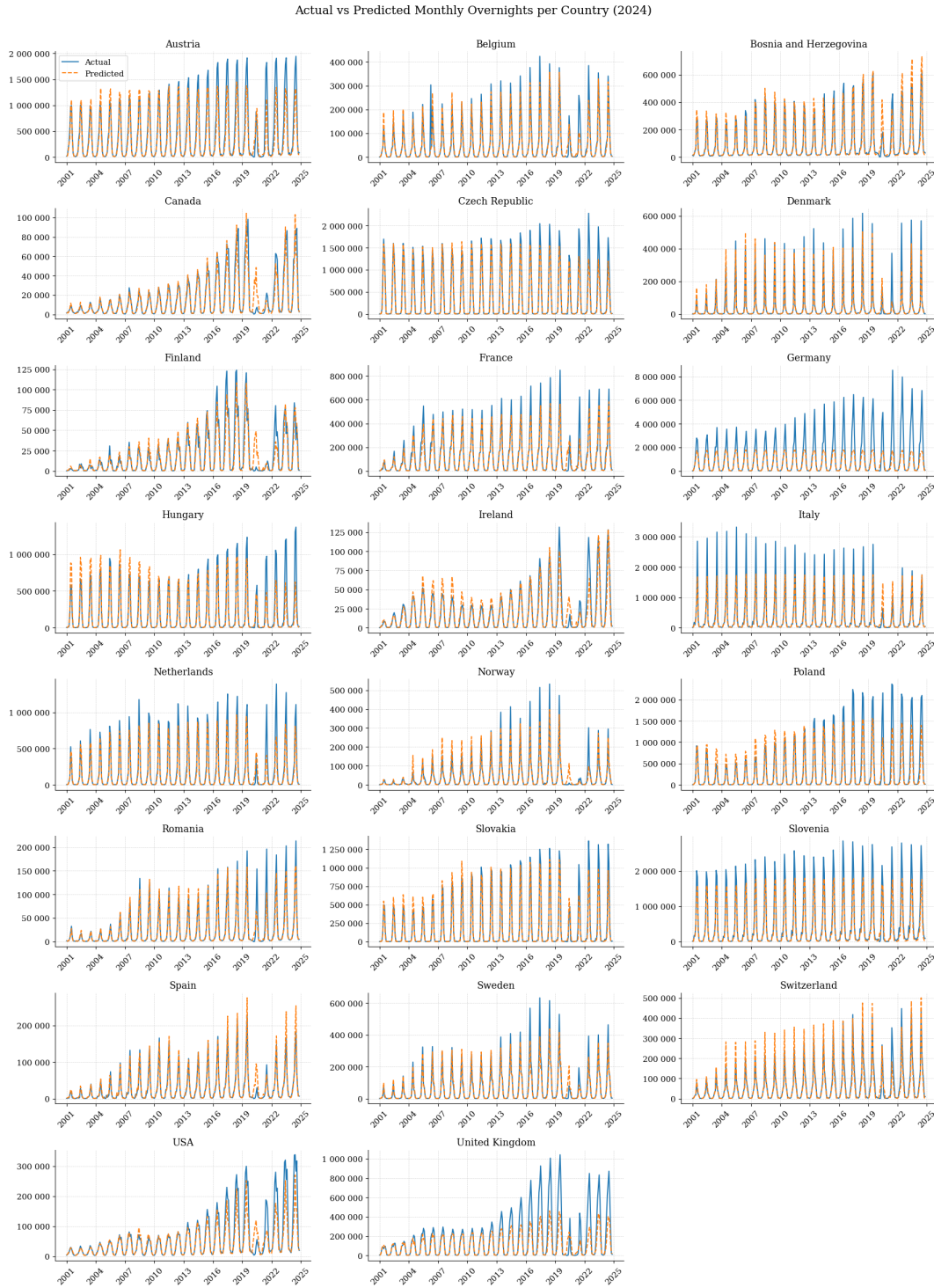


Figure 5.3: Comparison of actual and predicted total monthly overnights per country, full time series

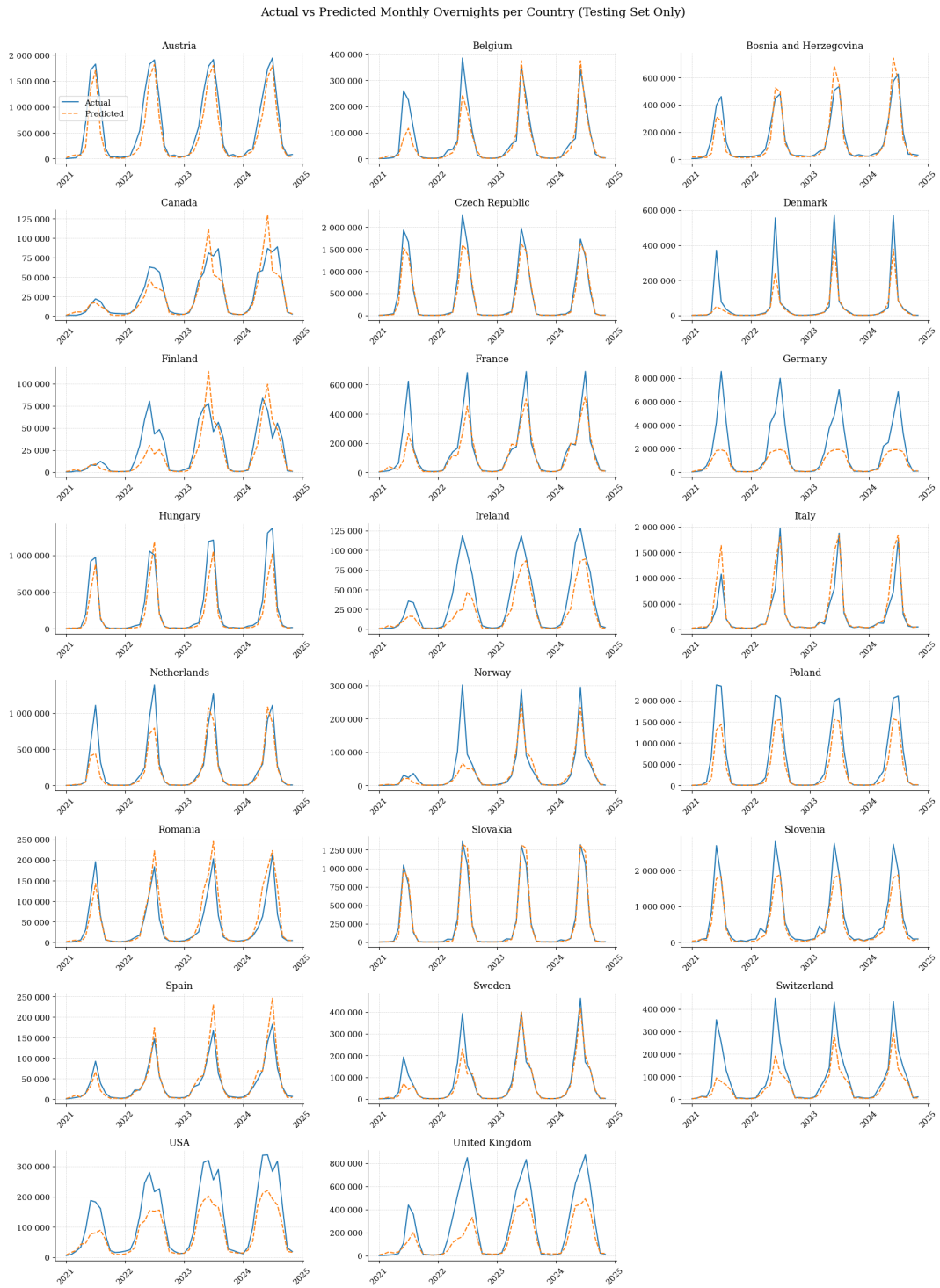


Figure 5.4: Comparison of actual and predicted total monthly overnights per country, testing set

To further investigate this phenomenon, a subset of countries was selected based on the countries with worst performances and a visual inspection of the results in Figure 5.3. The descriptive statistics summarized in Table A.1 con-

Table 5.2: ANN Model Performance by Country -- Overnights Forecast (2024) from predicted log-values

Country	MSE	RMSE	MAE	MAPE (%)	R ²
Italy	0.13	0.36	0.27	2.40	0.9377
Bosnia and Herzegovina	0.19	0.44	0.34	3.14	0.8850
Czech Republic	0.17	0.42	0.35	3.34	0.9669
Switzerland	0.23	0.48	0.39	3.77	0.9142
Netherlands	0.22	0.47	0.39	3.79	0.9519
Hungary	0.24	0.49	0.39	3.79	0.9335
Slovakia	0.30	0.55	0.38	3.84	0.9345
Poland	0.34	0.58	0.44	3.84	0.9329
Slovenia	0.35	0.59	0.47	4.01	0.8459
Austria	0.36	0.60	0.48	4.10	0.8653
France	0.31	0.56	0.44	4.19	0.8852
Canada	0.26	0.51	0.38	4.35	0.8702
Romania	0.29	0.54	0.43	4.48	0.8802
Sweden	0.32	0.57	0.42	4.51	0.9149
Belgium	0.32	0.57	0.43	4.73	0.9084
Germany	0.64	0.80	0.65	4.74	0.8444
Spain	0.27	0.52	0.44	4.78	0.8493
USA	0.41	0.64	0.56	4.89	0.7222
Denmark	0.38	0.61	0.47	5.18	0.9083
Norway	0.49	0.70	0.50	5.90	0.8753
United Kingdom	0.67	0.82	0.70	6.47	0.8218
Ireland	0.47	0.69	0.59	6.66	0.8558
Finland	0.58	0.76	0.58	7.10	0.8478

firm that each of the selected countries in this group has consistently high average values of arrivals and overnights. No other clear common factors or patterns were identified among this group that were not also present elsewhere in the dataset, suggesting that high tourist demand volume is the primary characteristic associated with model's underprediction in these cases. When the identical model architecture was applied exclusively to this subset of countries, the model was able to adjust more to the higher demand intensity and achieved slightly better predictive accuracy for some of the countries, as shown in Figure 5.5 and Table 5.3.

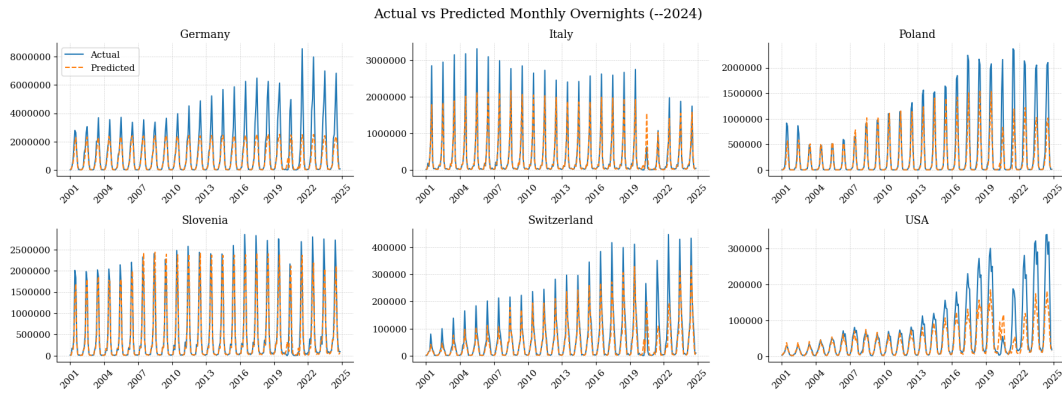


Figure 5.5: Comparison of actual and predicted monthly overnights trained only on high-volume countries, full dataset

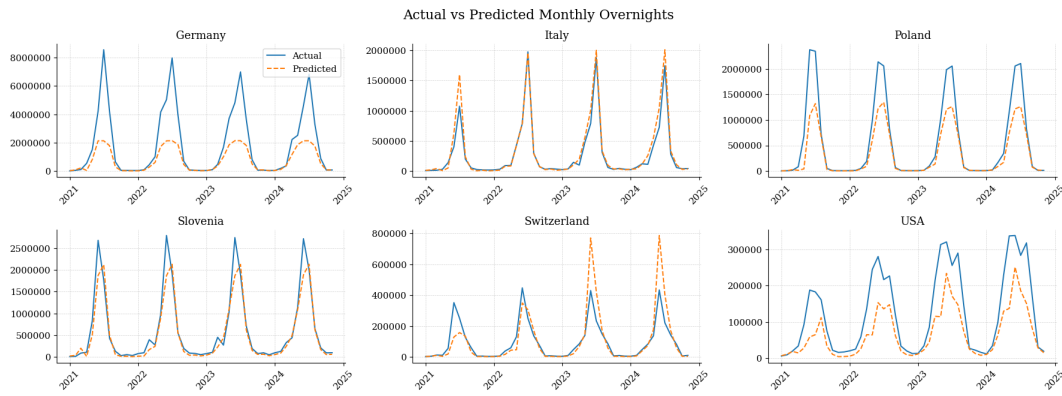


Figure 5.6: Actual and predicted monthly overnights trained on high-volume countries, testing set

The out-of-sample forecasts for USA, Slovenia and Germany had modest, but visible improvements when the mode was limited to high-volume subset of countries. In these cases, the model matches the actual values and shape of the testing set more closely. Poland had more limited gains in forecasting performance and Switzerland estimates overvalued the magnitude of post-pandemic recovery. It's possible that some of the macroeconomic indicators for Switzerland signaled this. Despite slight improvements in alignment during peak months, the model continues to underestimate the overall volumes and seasonal peaks. This suggests that while retraining on high-volume data slightly improves pattern recognition, it does not fully resolve the issue of scale underestimation in the most extreme cases. It is possible that additional model tuning, loss function weighting or even architecture modifications would be required to more effectively address large deviations in these particular markets.

Table 5.3: Model performance for subset of selected high-volume countries, Overnights Forecasts (2024)

Country	MSE	RMSE	MAE	MAPE (%)	R ²
Germany	0.3856	0.6210	0.4983	4.34	0.8710
Italy	0.2594	0.5093	0.3601	3.67	0.8254
Poland	0.3999	0.6324	0.3921	4.18	0.9024
Slovenia	0.5917	0.7691	0.5839	5.07	0.7401
Switzerland	0.1572	0.3965	0.2686	2.85	0.9246
USA	0.2852	0.5340	0.3916	4.20	0.8655

5.3 Key Relationships and Determinants of Tourism Demand

Permutation feature importance analysis highlights subtle but meaningful differences in the factors that are driving tourism demand in Croatia across the tested models. Seasonal patterns and autoregressive variables, particularly the 12-month lags consistently remain the dominant predictors across all tested models. However, the relative contribution of macroeconomic and policy variables, such as exchange rates, CPI, Schengen Area and EU Membership, Euro adoption, and Google Trends increases in models that include the pandemic and the subsequent recovery periods.

In models whose time series is restricted until 2019, nearly all predictive power is concentrated in temporal features. The dominance of seasonal and autoregressive components reflects the absence of any major economical shocks in this subset, as visible in Figure 5.7. As a result, the macroeconomic and policy indicators contribute less to the target variable forecasts, as presented in Figure 5.8 together with the derived permutation importance of each estimated model. One-hot-encoded months are collectively grouped into a single feature and represent calendar months, capturing the seasonal variation across all months in the year.

In the models estimated on the entire time series without the cutoff period and include the pandemic and post-pandemic years, we can observe a modest increase in the relevance of broader market signals such as exchange rates, euro adoption, and Google Trends. These shifts are small, but very likely reflect increased tourist sensitivity to economic uncertainty and external shocks. This suggests that during stable periods, tourism demand in Croatia is mainly shaped by regular seasonal cycles and momentum from previous months and years, while external factors are more relevant for anticipating and signaling

shocks or sudden changes in travel patterns.

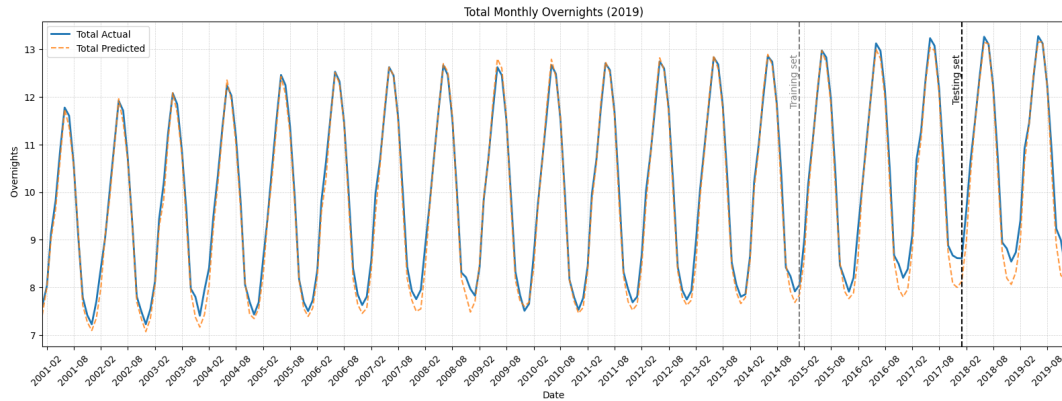


Figure 5.7: Forecasting performance of ANN Model for Overnights (log), 2019 cutoff

The thesis also presents permutation feature importance results from a separate model using only a subset of those countries that do not belong or have ever been part of Schengen Area. This is done as exploratory analysis to examine whether travel dynamics differ for markets outside the common visa area. In the model focused solely on non-Schengen countries, we observe a slightly different structure of predictive drivers. While seasonality and yearly lags remain dominant, the 1-month lag gains substantially more weight compared to other models, indicating that tourism demand from these countries is more sensitive to very recent trends or shocks. This may reflect differences in planning behavior, visa policies, or responsiveness to short-term travel conditions. Variables that represent the exchange rates and indicator on the adoption of Euro as a national currency retain an impact too, suggesting that market-related signals still play a role even outside the Schengen context. As expected, the indicator of Schengen Area membership has no explanatory power in this context, due to the applied filtering criteria.

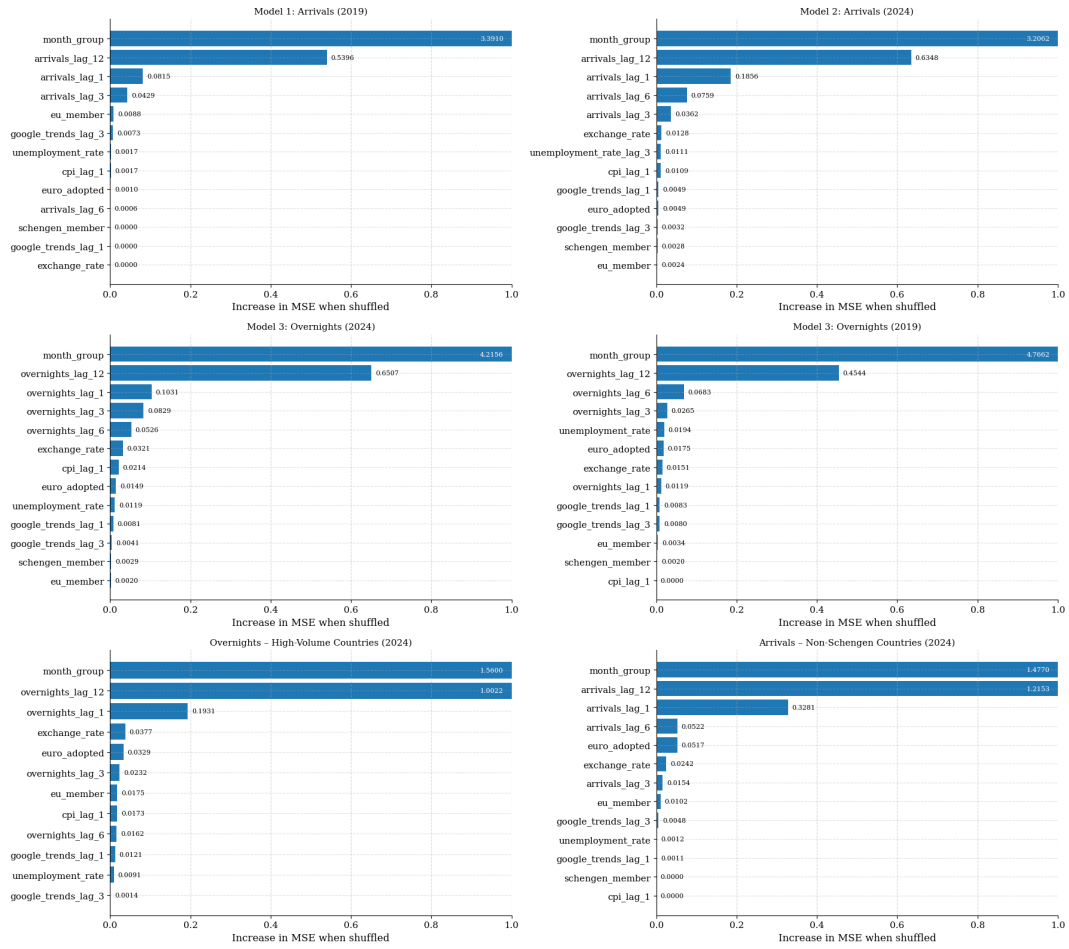


Figure 5.8: Permutation feature importance across models

Altogether, the results suggest that in stable periods, tourism demand tends to be slightly more driven by predictable seasonal and historical trends. By contrast, during more volatile periods, results point to external signals playing a larger role in anticipating deviations and detecting changes in travel patterns. Overall, while seasonality and past demand remain foundational to forecasting, external variables can play a more active role in times of volatility, helping to signal shocks and improve model robustness during more turbulent periods.

Chapter 6

Conclusion

This thesis investigated tourism demand in Croatia and its forecasting possibilities, using monthly data on tourist arrivals and overnights from 23 countries of origin. By analyzing a comprehensive dataset spanning the past 25 years and comparing neural network forecasts against a traditional autoregressive benchmark model, the study identified the factors which most influence tourism demand and how these patterns evolve over time and across different countries. The comparative analysis of several neural network model structures brought important findings. A major part of the research compared model performance using the full time series dataset with models that were restricted until the year 2019 and only used data that was isolated from the COVID-19 pandemic and its subsequent recovery years. The results showed that when this period of economic shock was included, it improved the model's ability to predict future tourism demand, as the models could learn from both normal conditions and periods of major disruptions. Models trained on the complete dataset demonstrated better generalization and predictive accuracy than those limited to pre-pandemic data. This suggests that exposure to economic shocks enables the model to learn more robust patterns for forecasting future demand.

Contrary to some previous research, specifically Claveria *et al.* (2015), the results indicated that models targeting overnights achieved slightly higher predictive accuracy than those forecasting tourist arrivals. This may reflect the fact that overnights more closely capture trip duration and overall tourism impact, offering a more stable basis for demand prediction in the context of Croatia.

While the benchmark AR models performed relatively well in stable periods before the pandemic, they showed significantly weaker results when applied to

the full dataset. In particular, their performance deteriorated in the presence of external shocks such as COVID-19, with considerably higher error metrics and lower explanatory power. In contrast, forecasts done by artificial neural networks consistently achieved stronger predictive accuracy across all time periods and target variables. This suggests that, while AR models may still be a valid option for short-term forecasting in predictable conditions, ANNs offer a more flexible framework for modeling complex and real-world patterns, especially when structural changes or disruptions are present in the data.

Per-country analysis revealed that the neural networks can effectively capture trends and seasonality for most origin countries. However, the models exhibited a tendency to underpredict demand for countries with exceptionally high tourism volumes, especially during periods of major disruption. This limitation likely arises from both the effect of the log transformation and the model's central tendency in the presence of extreme values. Conducting a targeted analysis on high-volume countries helped to address this challenge and resulted in slightly improved forecasting accuracy for these countries.

The analysis also revealed that the factors influencing tourism demand can shift moderately depending on the external conditions. In all models, seasonality indicators and autoregressive features capturing recent trends in arrivals and overnights consistently show to be the most important predictors of tourism demand. This pattern is especially pronounced in the pre-pandemic period, where tourism flows followed stable, predictable cycles. It also explains why the benchmark AR(12) model performed comparatively well during those years, relying solely on the persistence of past values to forecast future demand. However, when the models accounted for the pandemic and the recovery years, economic variables such as exchange rates, consumer prices, and online search interest became slightly more important. This suggests that while Croatian tourism demand is largely shaped by seasonal cycles and past values under normal conditions, its tourism demand can become more sensitive to broader economic factors during periods of crisis or significant change.

In summary, this thesis demonstrates that forecasting tourism demand in Croatia is most effective when models combine information on seasonal cycles, recent trends, and relevant economic factors. By comparing different neural network setups, time periods, and traditional AR benchmarks, the research shows that neural networks are well-equipped to capture both predictable patterns and unexpected shifts in tourism flows. These results provide useful insights for improving tourism demand forecasts in Croatia and may help inform

future planning and strategy for tourism authorities and businesses. However, some limitations should be acknowledged. The findings rely on historical data, which may not fully reflect future developments or structural changes in the tourism industry. The use of log transformation, while helpful for stabilizing variance, can limit the model's responsiveness to extreme values, especially for high-volume countries. In addition, certain qualitative factors, such as abrupt policy changes or geopolitical events, are not explicitly modeled. For future researchers, it could be recommended to integrate more frequent and timely data sources, such as real-time mobility or sentiment indicators. Furthermore, grouping countries of origin based on shared characteristics, such as market size, economic ties, or travel behavior and developing tailored forecasting models for these groups could help address persistent underprediction in high-volume markets and enhance overall forecasting accuracy. Finally, explicitly accounting for major events or policy interventions may support even more resilient planning in the tourism sector.

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Appendix A

EDA

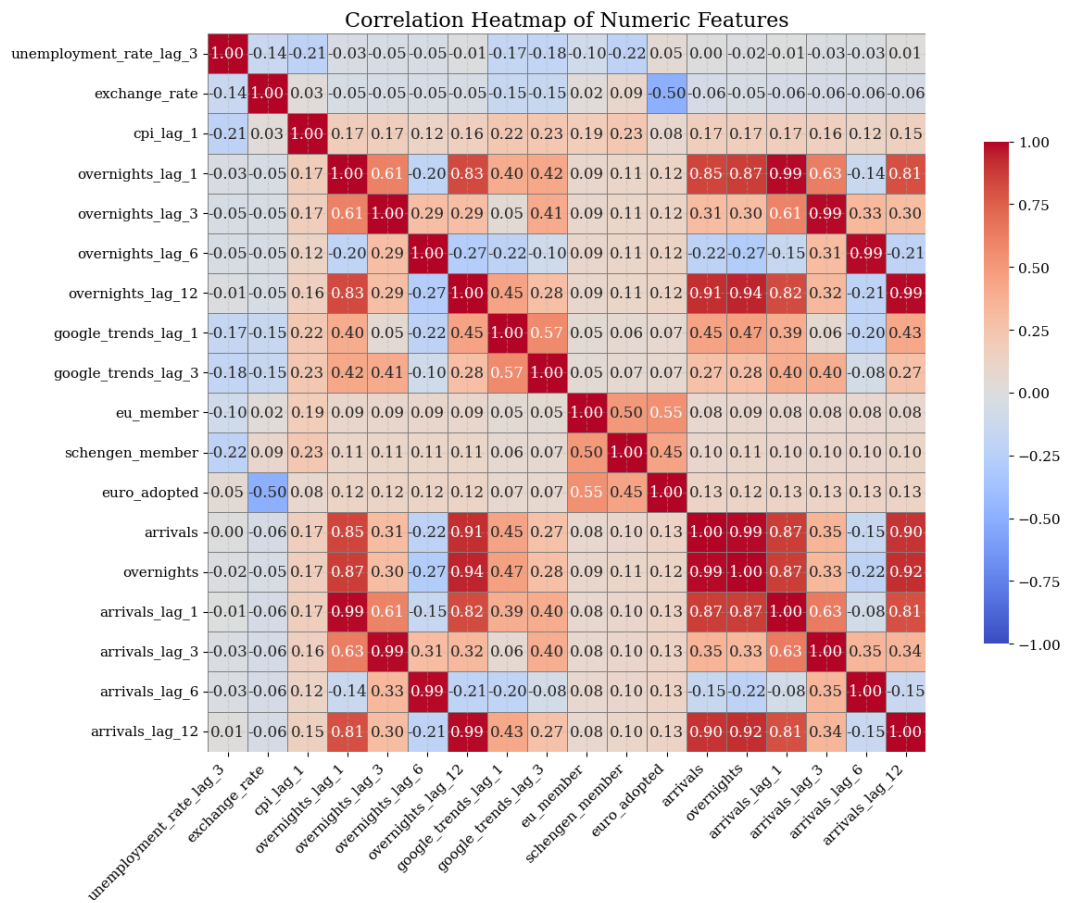


Figure A.1: Pearson correlation matrix

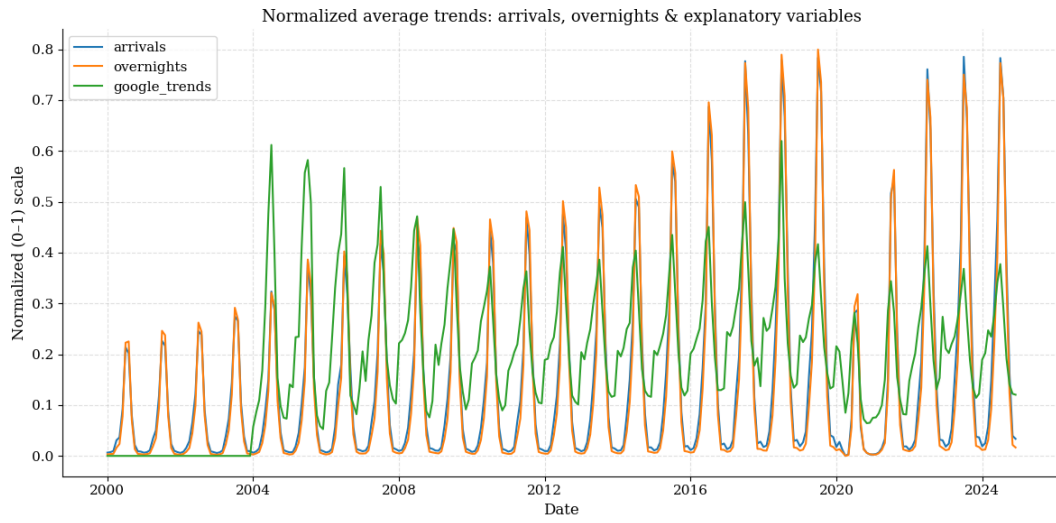


Figure A.2: Average normalized trends of Arrivals, Overnights and Google Trends data

Table A.1: Descriptive statistics for the selected subset of countries with high-volume tourist demand

Variable	Czech Rep.	Germany	Italy	Poland	Slovenia	USA	UK	Dataset
<i>Arrivals</i>								
mean	9.44	11.33	10.33	9.84	11.04	9.93	9.66	8.79
std	2.07	1.84	1.29	2.11	1.33	1.54	1.95	2.04
<i>Overnights</i>								
mean	10.93	12.95	11.46	11.35	12.30	11.11	11.11	10.03
std	2.35	2.11	1.49	2.32	1.62	1.25	2.00	2.24
<i>Exchange Rate</i>								
mean	3.25	0.69	0.69	1.71	0.69	0.74	0.62	1.34
std	0.03	0.00	0.00	0.03	0.00	0.03	0.01	1.18
<i>Google Trends</i>								
mean	10.17	33.17	8.68	11.81	14.02	38.30	30.91	19.97
std	6.07	13.73	5.32	6.57	6.66	11.27	14.40	19.04
<i>Unemployment Rate</i>								
mean	2.65	3.27	7.98	3.00	4.06	4.17	4.17	7.85
std	0.34	0.34	1.29	0.37	0.62	0.92	0.35	4.83
<i>CPI</i>								
mean	4.99	4.85	4.82	4.99	4.84	4.91	4.93	4.65
std	0.11	0.06	0.06	0.11	0.07	0.06	0.07	0.16

Appendix B

Forecasting Performance

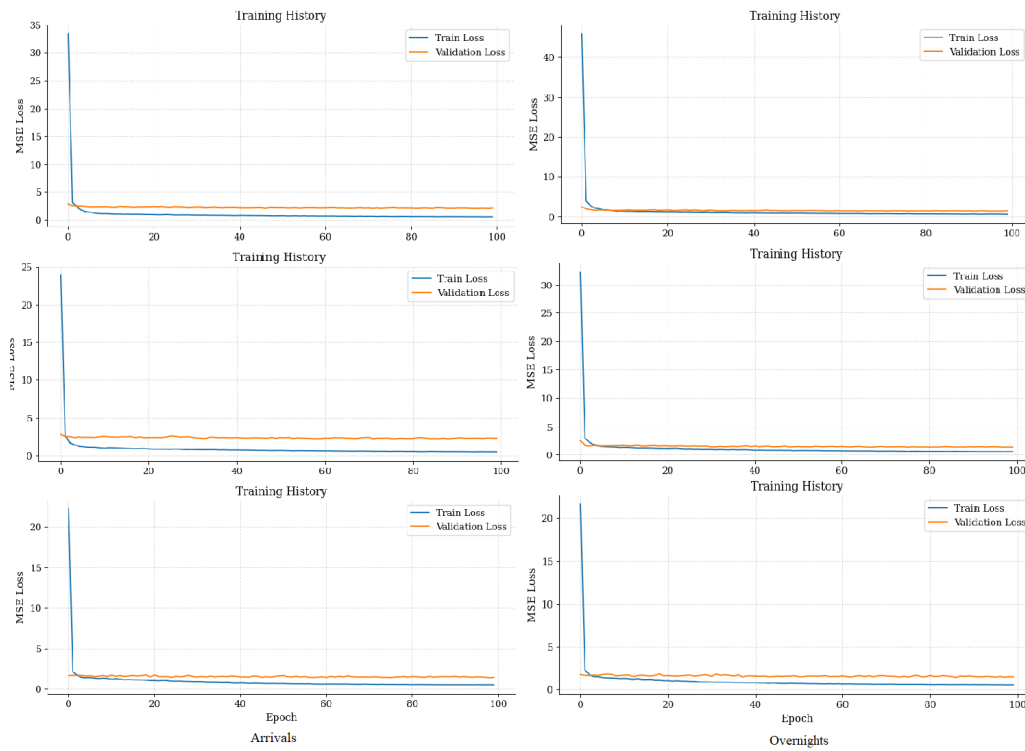


Figure B.1: Training loss (MSE) curve across estimated ANN models with learning rates 0.002, 0.003 and 0.005, respectively

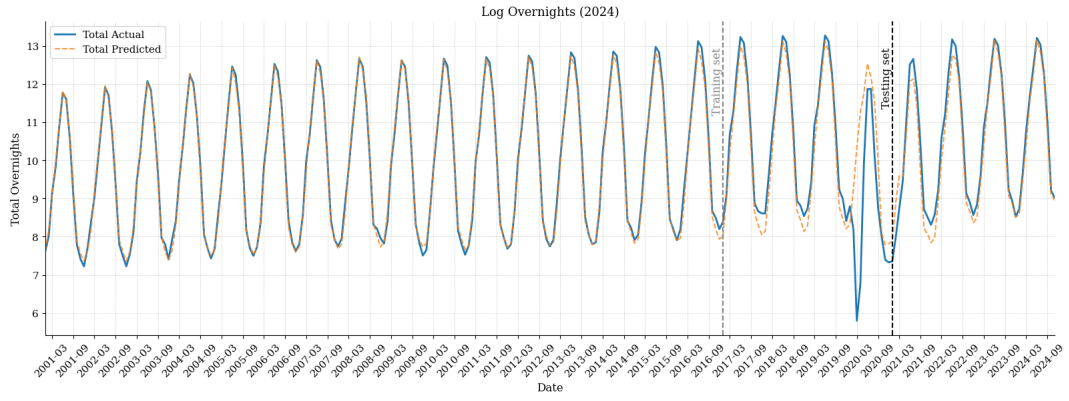


Figure B.2: Forecasting performance of ANN Model for Total Overnights (log), full dataset

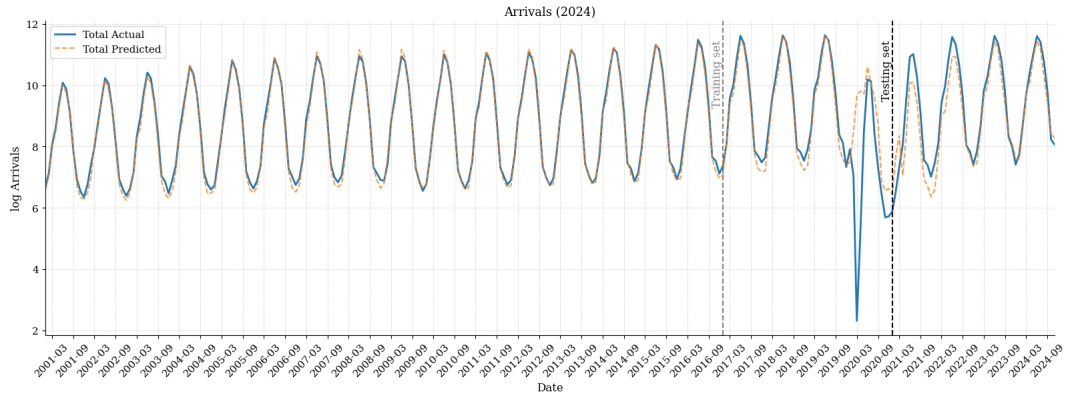


Figure B.3: Forecasting performance of ANN Model for Total Arrivals (log), full dataset

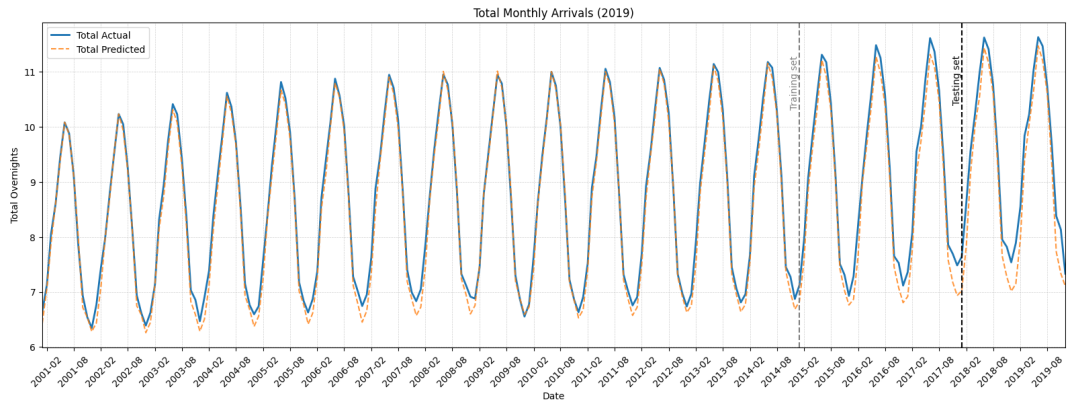


Figure B.4: Forecasting performance of ANN Model for Total Arrivals (log), 2019 cutoff

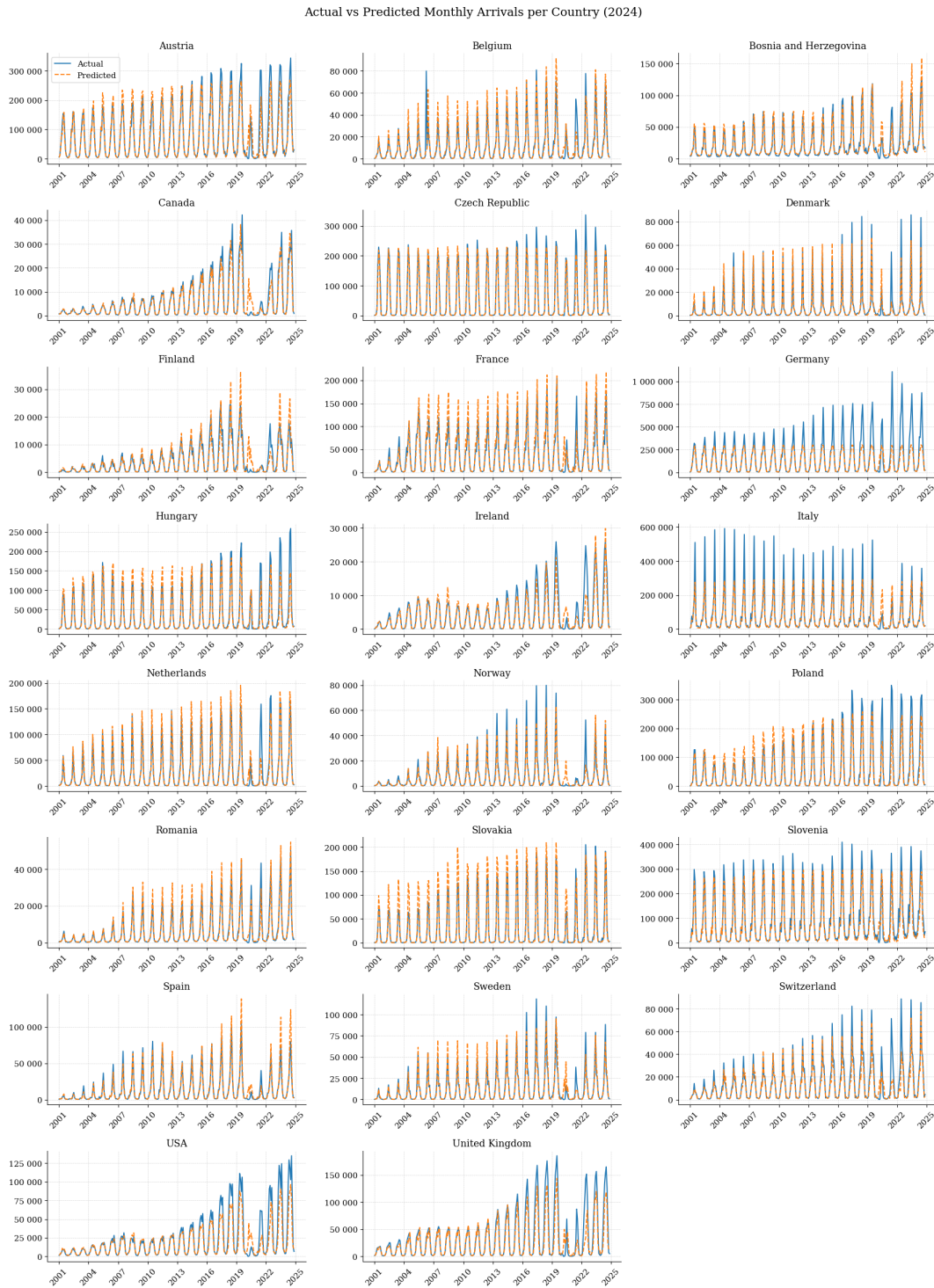


Figure B.5: Forecasting performance of ANN Model for Arrivals per country, full dataset

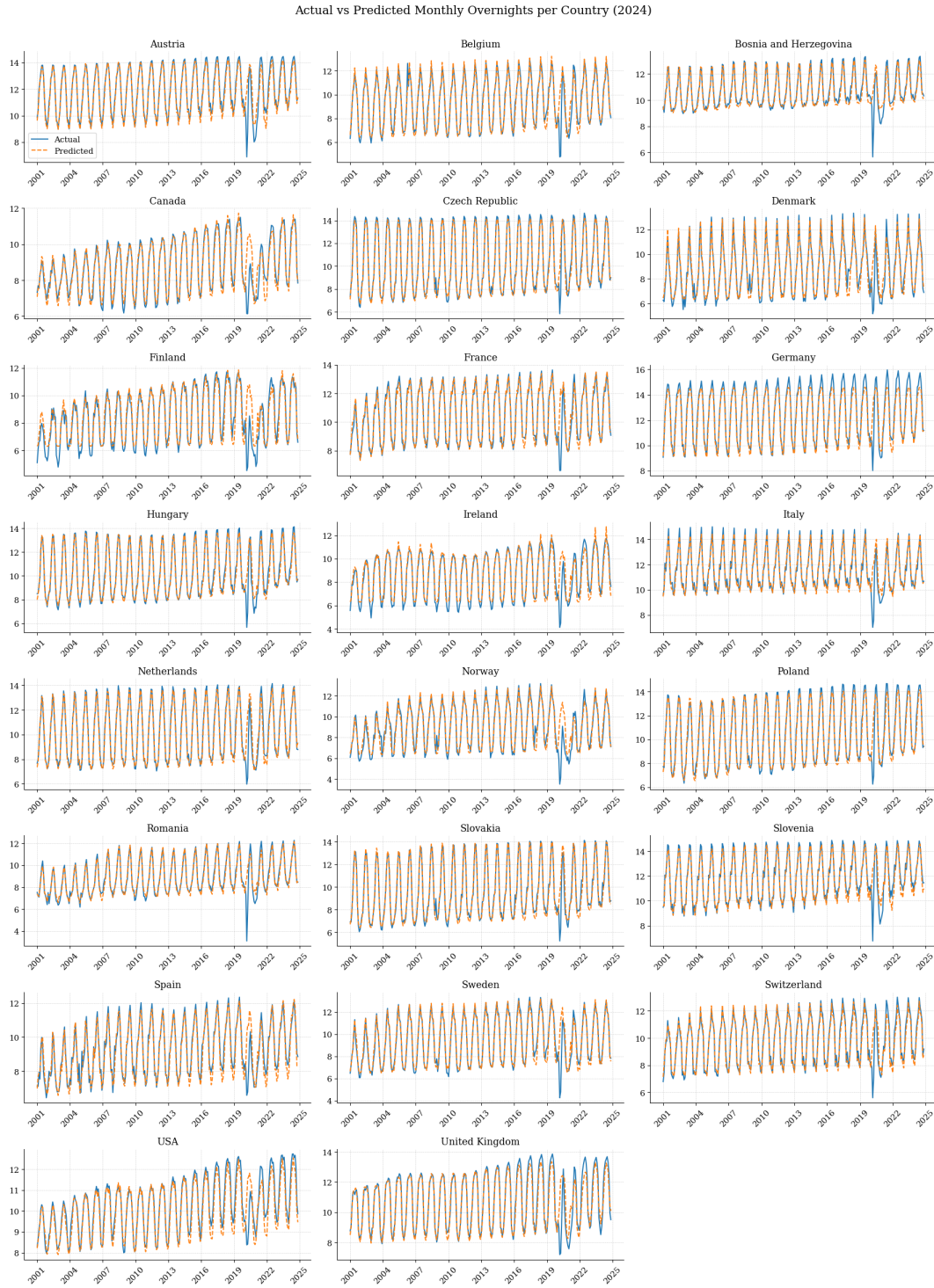


Figure B.6: Forecasting performance of ANN Model for Overnights (log) per country, full dataset

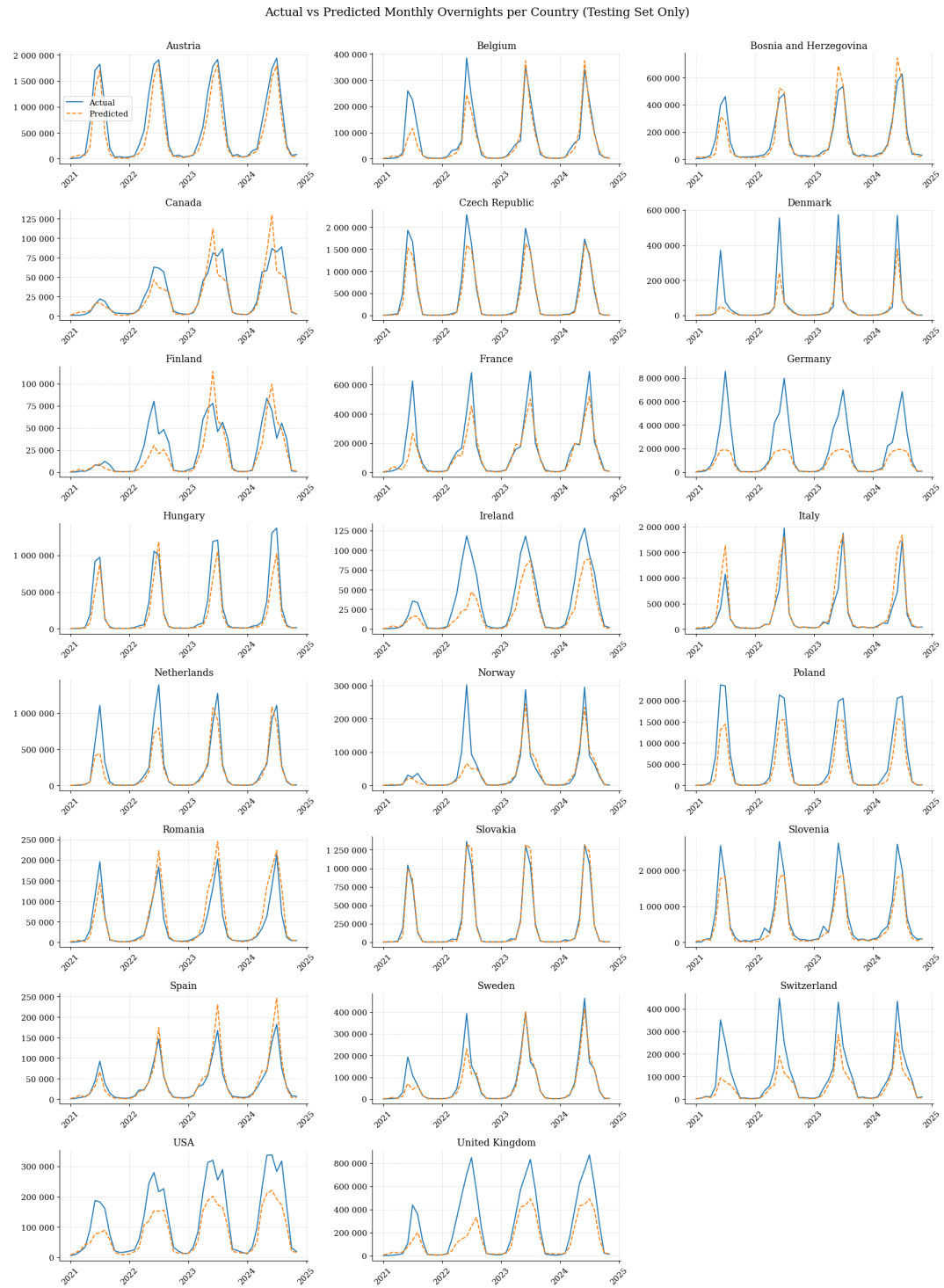


Figure B.7: Forecasting performance of ANN Model for Overnights (log) per country, testing set

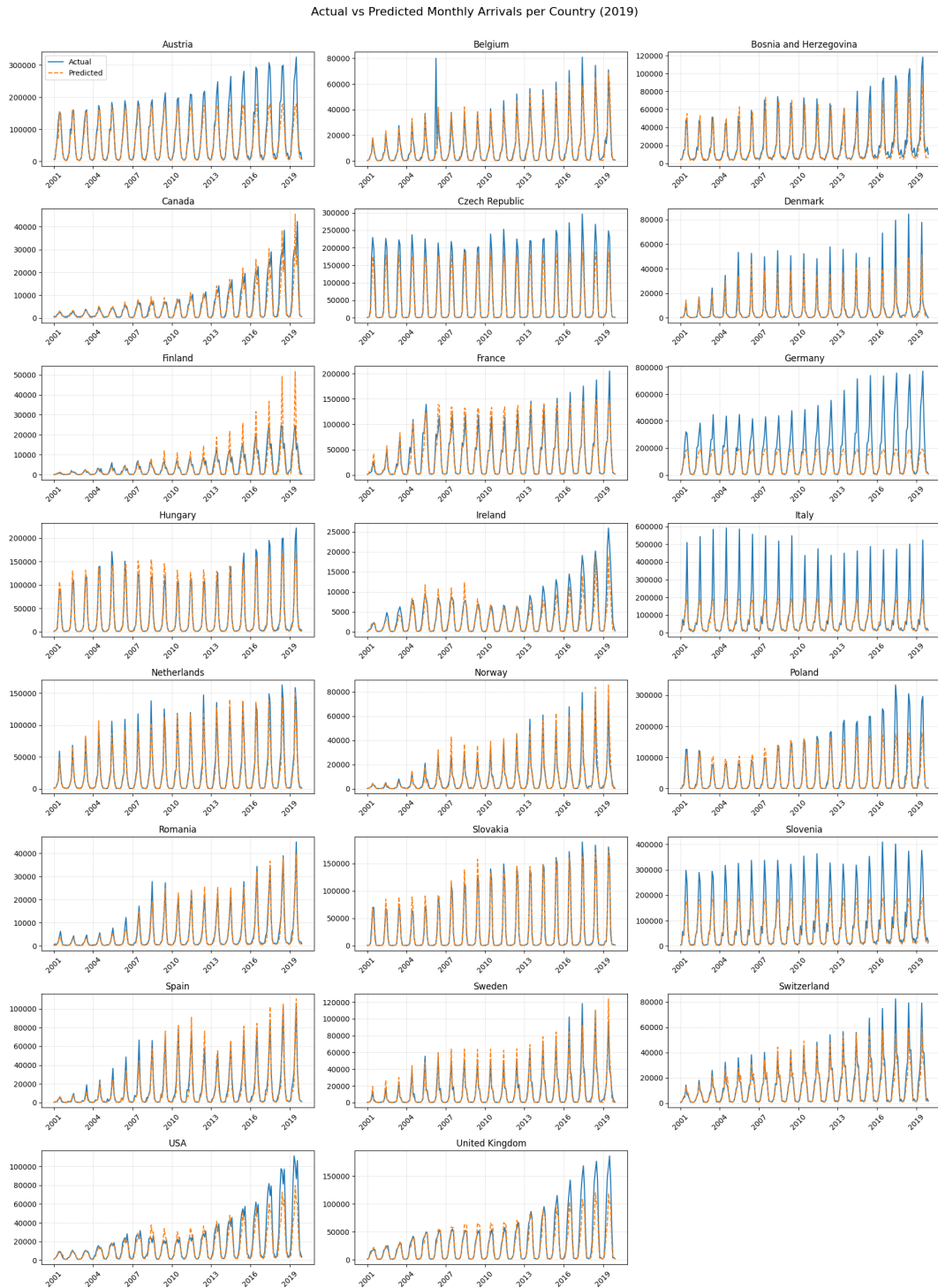


Figure B.8: Forecasting performance of ANN Model for Arrivals per country, 2019 cutoff

Table B.1: Monthly predictions of overnight stays: Neural Network vs. AR(12)

Date	Act	ANN	AR(12)	Act (log)	ANN (log)	AR (log)
2021-01-01	62 466	142 963	303 287	11.042	11.870	12.3430
2021-02-01	150 716	328 693	600 796	11.923	12.703	12.6224
2021-03-01	350 199	700 691	2 316 442	12.766	13.460	13.3060
2021-04-01	1 087 846	848 211	4 662 807	13.900	13.651	14.6555
2021-05-01	5 310 517	2 647 898	13 367 747	15.485	14.789	15.3551
2021-06-01	18 285 648	9 663 047	23 864 606	16.722	16.084	16.4084
2021-07-01	22 835 797	12 279 330	26 766 090	16.944	16.323	16.9879
2021-08-01	9 089 504	5 162 002	10 389 479	16.023	15.457	17.1026
2021-09-01	1 655 190	947 237	2 330 979	14.319	13.761	16.1563
2021-10-01	253 002	148 680	550 319	12.441	11.910	14.6618
2021-11-01	248 039	168 986	410 243	12.421	12.038	13.2183
2021-12-01	194 665	107 331	410 243	12.179	11.584	12.9245
2022-01-01	269 303	141 455	245 558	12.504	11.860	12.4113
2022-02-01	458 095	306 375	323 832	13.035	12.633	12.6880
2022-03-01	1 894 671	1 156 249	658 249	14.455	13.961	13.3973
2022-04-01	3 402 278	2 195 879	2 398 252	15.040	14.602	14.6903
2022-05-01	11 530 088	6 004 606	5 152 017	16.260	15.608	15.4549
2022-06-01	22 622 422	13 874 254	14 403 131	16.934	16.446	16.4830
2022-07-01	24 451 654	15 520 567	24 990 627	17.012	16.558	17.0340
2022-08-01	9 960 236	7 216 565	28 217 199	16.114	15.792	17.1554
2022-09-01	2 149 740	1 617 536	10 947 091	14.581	14.296	16.2086
2022-10-01	410 559	303 037	2 471 588	12.925	12.622	14.7204
2022-11-01	379 666	290 437	638 257	12.847	12.579	13.3665
2022-12-01	254 995	238 902	423 811	12.449	12.384	12.9570
2023-01-01	296 865	267 457	263 912	12.601	12.497	12.4834
2023-02-01	554 437	505 745	345 851	13.226	13.134	12.7538
2023-03-01	2 252 849	1 533 497	719 235	14.628	14.243	13.4859
2023-04-01	4 574 565	3 047 959	2 499 844	15.336	14.930	14.7317
2023-05-01	11 448 105	7 864 847	5 688 769	16.253	15.878	15.5540
2023-06-01	21 970 197	16 584 534	15 441 693	16.905	16.624	16.5526
2023-07-01	23 403 745	16 560 836	26 433 075	16.968	16.623	17.0901
2023-08-01	10 123 178	7 749 106	29 712 696	16.130	15.863	17.2071
2023-09-01	2 370 800	1 735 212	11 502 209	14.679	14.367	16.2580
2023-10-01	396 749	362 384	2 657 118	12.891	12.800	14.7928
2023-11-01	428 790	357 101	728 314	12.969	12.786	13.4985

Date	Act	ANN	AR(12)	Act (log)	ANN (log)	AR (log)
2023-12-01	252 219	294 189	442 448	12.438	12.592	13.0001
2024-01-01	317 259	296 746	284 156	12.667	12.601	12.5573
2024-02-01	827 374	527 448	370 009	13.626	13.176	12.8213
2024-03-01	1 987 342	1 622 948	783 740	14.502	14.300	13.5718
2024-04-01	5 732 740	3 179 554	2 623 577	15.562	14.972	14.7800
2024-05-01	10 660 914	8 056 109	6 267 838	16.182	15.902	15.6509
2024-06-01	21 845 134	17 038 892	16 524 240	16.899	16.651	16.6203
2024-07-01	23 358 729	16 478 566	28 170 177	16.966	16.618	17.1538
2024-08-01	9 622 687	7 826 359	31 235 866	16.080	15.873	17.2571
2024-09-01	2 597 620	1 839 033	12 097 888	14.770	14.425	16.3085
2024-10-01	456 947	378 064	2 880 537	13.032	12.843	14.8735
2024-11-01	449 835	372 818	821 457	13.017	12.829	13.6188